

A Numerical Method for Exact Boundary Controllability Problems for the Wave Equation

M. GUNZBURGER

School of Computational Science, Florida State University
Tallahassee, FL 32306-4120, U.S.A.
gunzburg@csit.fsu.edu

L. S. HOU

Department of Mathematics, Iowa State University
Ames, IA 50011-2064, U.S.A.
hou@math.iastate.edu

L. JU

Department of Mathematics, University of South Carolina
Columbia, SC 29208, U.S.A.
ju@math.sc.edu

Abstract—The computational approximation of exact boundary controllability problems for the wave equation in two dimensions is studied. A numerical method is defined that is based on the direct solution of optimization problems that are introduced in order to determine unique solutions of the controllability problem. The uniqueness of the discrete finite-difference solutions obtained in this manner is demonstrated. The convergence properties of the method are illustrated through computational experiments. Efficient implementation strategies for the method are also discussed. It is shown that for smooth, minimum L^2 -norm Dirichlet controls, the method results in convergent approximations without the need to introduce regularization. Furthermore, for the generic case of nonsmooth Dirichlet controls, convergence with respect to L^2 norms is also numerically demonstrated. One of the strengths of the method is the flexibility it allows for treating other controls and other minimization criteria; such generalizations are discussed. In particular, the minimum H^1 -norm Dirichlet controllability problem is approximated and solved, as are minimum regularized L^2 -norm Dirichlet controllability problems with small penalty constants. Finally, a discussion is provided about the differences between our method and existing methods; these differences may explain why our methods provide convergent approximations for problems for which existing methods produce divergent approximations unless they are regularized in some manner. © 2006 Elsevier Ltd. All rights reserved.

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1. INTRODUCTION

Given a positive number T and a bounded domain $\Omega \in \mathbb{R}^N$ with boundary Γ , let $Q = \Omega \times (0, T)$ and $\Sigma = \Gamma \times (0, T)$ denote a space-time cylinder and its lateral surface, respectively. The *general exact boundary controllability problem for the wave equation* is defined as follows. Given functions $u_0(\mathbf{x})$ and $u_1(\mathbf{x})$ defined on Ω , find a state function u defined on Q and a control

function g defined on Σ such that

$$\text{the linear wave equation} \quad u_{tt} - \Delta u = 0, \quad \text{in } Q, \quad (1)$$

$$\text{the initial conditions} \quad u|_{t=0} = u_0, \quad \text{and} \quad u_t|_{t=0} = u_1, \quad \text{in } \Omega, \quad (2)$$

$$\text{the terminal conditions} \quad u|_{t=T} = 0, \quad \text{and} \quad u_t|_{t=T} = 0, \quad \text{in } \Omega, \quad (3)$$

$$\text{the boundary condition} \quad B(u) = g, \quad \text{on } \Sigma, \quad (4)$$

are satisfied. In (4), $B(\cdot)$ is a boundary operator which in general may be linear or nonlinear and may or may not involve derivatives of u . It is also possible that $B(\cdot)$ takes different forms on different portions of the lateral boundary Σ . Of course, the operator $B(\cdot)$ is determined by how, in the physical problem that is being modeled, one wants to apply control. Control could also be applied on only a subset of the boundary and all of our deliberations can be extended to this case; however, for the sake of simplicity, we will only discuss the case wherein control is applied on all of the boundary, i.e., as in (4).

In general, solutions of the exact controllability problem (1)–(4), when they exist, are not unique. A particular solution from the set of solutions (if such a set is not empty) can be extracted by solving the optimization problem

$$\min_{u \in G} \mathcal{J}(u), \quad \text{subject to (1)–(4)}, \quad (5)$$

where $\mathcal{J}(\cdot)$ is a given quadratic, positive functional and G an appropriate function space. The functional $\mathcal{J}(\cdot)$ should also be determined from the physical problem being modeled.

The new computational methods discussed in this paper apply to the general exact boundary controllability problem (1)–(5). In fact, the methods will actually involve the solution of discretized versions of the problem

$$\min_{u \in G} \mathcal{J}(u), \quad \text{subject to (1)–(3)}, \quad (6)$$

for which *no boundary conditions are applied*. Once a solution u of problem (6) is obtained, one may subsequently determine a control g from (4), i.e., by setting $g = B(u)$ for any given appropriate operator $B(\cdot)$ mapping functions defined on Q to functions defined on Σ . (Of course, in practice, this is all done after discretization.) Thus our methods are general in a number of respects:

- one may choose an “arbitrary” functional $\mathcal{J}(\cdot)$ in the minimization problem (6) which is invoked in order to extract a particular solution from the set of solutions of (1)–(5);
- since no boundary conditions are imposed in the problem we solve, i.e., see (6), one may, *after the solution is obtained*, choose an “arbitrary” control function g by evaluating $g = B(u)$ for an “arbitrary” operator $B(\cdot)$;
- again, since no boundary conditions are imposed in the problem we solve, a single solution of that problem can be used to define multiple optimal controls simply by choosing different operators $B(\cdot)$.

The literature, however, is almost exclusively devoted to a *single* special case of the general exact boundary controllability problem (5). First, the boundary operator $B(\cdot)$ is chosen to be the trace operator which restricts u defined on Q to Σ so that (4) merely becomes a Dirichlet boundary condition. Second, the functional in (5) is chosen to be the square of the $L^2(\Sigma)$ -norm of the solution u . Since it is also required that $u|_{\Sigma} = g$, this is equivalent to minimizing the $L^2(\Sigma)$ -norm of the control. We will study this special case, but we will consider other settings as well; see Sections 1.1 and 1.2 for descriptions of the specific controllability problems that we focus on.

1.1. The Minimum $L^2(\Sigma)$ -Norm Exact Dirichlet Boundary Controllability Problem

Given functions $u_0(\mathbf{x})$ and $u_1(\mathbf{x})$ defined on Ω , find a state function u defined on Q and a control function g defined on Σ such that (1)–(3) and

$$\text{the boundary condition,} \quad u|_{\Sigma} = g, \quad (7)$$

are satisfied. It is well known that state and control functions u and g such that (1)–(3) and (7) are satisfied exist provided T is sufficiently large; see, e.g., [1–4]. This result is not surprising; equation (1) describes the propagation of waves with speed $c = 1$ and one can expect that exact boundary controllability is realized if $T \geq \sup_{\mathbf{x}, \mathbf{y} \in \Omega} d(\mathbf{x}, \mathbf{y})/c = \sup_{\mathbf{x}, \mathbf{y} \in \Omega} d(\mathbf{x}, \mathbf{y})$, where $d(\mathbf{x}, \mathbf{y})$ denotes the standard Euclidian distance between $\mathbf{x}$ and $\mathbf{y}$ in $\mathbb{R}^N$. This prediction is supported by theory [1,3] and computational experiments [5,6]. It is worth noting that a computational solution may be obtained for any value of T since finite-dimensional systems are in general exactly controllable [7]; however, for values of T that are too small, the computational solutions will not converge as the temporal and spatial grid sizes tend to zero.

In general, solutions of the exact Dirichlet boundary controllability problem (1)–(3) and (7), when they exist, are not unique. However, using the *Hilbert uniqueness method* (HUM), it was shown in [1,2] that a solution of the exact Dirichlet boundary controllability problem having minimum $L^2(\Sigma)$ -norm is unique. (Throughout, for $s \in \mathbb{R}$, $H^s(\cdot)$ denotes the standard Sobolev space; see, e.g., [8].)

PROPOSITION 1.1. *Let $u_0 \in L^2(\Omega)$ and $u_1 \in H^{-1}(\Omega)$ be given. Let*

$$\mathcal{J}_0(u) = \|u\|_{L^2(\Sigma)}^2 \equiv \int_{\Sigma} u^2 d\Gamma dt. \quad (8)$$

Then, if T is sufficiently large, the optimization problem

$$\min_{u|_{\Sigma} \in L^2(\Sigma)} \mathcal{J}_0(u), \quad \text{subject to (1)–(3) and (7),} \quad (9)$$

has a unique solution.

PROOF. See [1].

The controllability problem (9) is the special case that is considered in most of the literature; it chooses the control to be of Dirichlet type and extracts a unique solution from the set of solutions by requiring the minimization of the $L^2(\Sigma)$ -norm of the solution. Note again that in our methods we will not actually need to apply the boundary condition (7) when we solve the minimization problem (9).

1.2. Minimum $H^1(\Sigma)$ -Norm and

Minimum Regularized $L^2(\Sigma)$ -Norm Exact Boundary Controllability Problems

For $0 \leq \epsilon \leq 1$, let

$$\mathcal{J}_{\epsilon}(u) = \|u\|_{L^2(\Sigma)}^2 + \epsilon \|u\|_{H^1(\Sigma)}^2, \quad \text{where } \|u\|_{H^1(\Sigma)}^2 \equiv \int_{\Sigma} (|u_t|^2 + \nabla_s u \cdot \nabla_s u) d\Gamma dt, \quad (10)$$

with ∇_s denoting the gradient operator acting tangentially to the boundary Γ . Note that if $\epsilon = 0$, we recover the functional in (8) and that $\mathcal{J}_1(\cdot) = \|\cdot\|_{H^1(\Sigma)}^2$. Then, we will also consider the *minimum $H^1(\Sigma)$ -norm exact boundary controllability problem*

$$\min_{u|_{\Sigma} \in H^1(\Sigma)} \mathcal{J}_1(u), \quad \text{subject to (1)–(3),} \quad (11)$$

and, for $0 < \epsilon < 1$, the *minimum regularized $L^2(\Sigma)$ -norm exact boundary controllability problem*

$$\min_{u|_{\Sigma} \in H^1(\Sigma)} \mathcal{J}_{\epsilon}(u), \quad \text{subject to (1)–(3).} \quad (12)$$

In particular, we will study the Dirichlet control version of these problems, i.e., problems in which control is applied through (7).

1.3. Previous Computational Approaches for the Minimum $L^2(\Sigma)$ -Norm Exact Dirichlet Controllability Problem

In this section, we briefly review existing computational approaches for the special case (9).

HUM-based approach

The Hilbert uniqueness method (HUM) approach provides a systematic way to construct a control g that solves the controllability problem (9), i.e., of determining a minimum $L^2(\Sigma)$ -norm solution of the exact Dirichlet boundary controllability problem. In the sequel, $H_0^1(\Omega) = \{v \in H^1(\Omega) \mid v|_\Gamma = 0\}$.

Let $E = H_0^1(\Omega) \times L_2(\Omega)$ and $F = H^{-1}(\Omega) \times L_2(\Omega)$. For any $(e_0, e_1) \in E$, solve the forward-in-time linear wave equation problem

$$\begin{aligned} \phi_{tt} - \Delta \phi &= 0, & \text{in } Q, \\ \phi &= 0, & \text{on } \Sigma, \\ \phi|_{t=0} &= e_0, \quad \phi_t|_{t=0} = e_1, & \text{in } \Omega, \end{aligned} \quad (13)$$

and then solve the backward-in-time linear wave equation problem

$$\begin{aligned} \psi_{tt} - \Delta \psi &= 0, & \text{in } Q, \\ \psi &= \partial_n \phi, & \text{on } \Sigma, \\ \psi|_{t=T} &= 0, \quad \psi_t|_{t=T} = 0, & \text{in } \Omega. \end{aligned} \quad (14)$$

This enables us to (implicitly) define a linear operator $\Lambda : E \rightarrow F$ by

$$\Lambda(e_0, e_1) = (\psi_t|_{t=0}, -\psi|_{t=0}). \quad (15)$$

It is known (see [1,2]) that Λ is an isomorphism from E to F if T is sufficiently large so that one may solve the problem of finding (e_0, e_1) such that

$$\Lambda(e_0, e_1) = (u_1, -u_0). \quad (16)$$

For this choice of e_0 and e_1 , it is known (again see [1,2]) that $\psi(\mathbf{x}, t)$ provides the unique solution of the minimum $L^2(\Sigma)$ -norm controllability problem (9). The corresponding unique minimum $L^2(\Sigma)$ -norm control is obtained by setting $g = \psi|_\Sigma = (\partial_n \phi)|_\Sigma$.

It is also known (see [1]) that if T is sufficiently large, Λ is a strongly elliptic, self-adjoint operator. Based on these observations, a conjugate gradient method was developed in [1,5,6,9] to solve (16). The numerical implementation of the method was effected through a combination of a finite-element method for the spatial discretization and a finite-difference method for the temporal discretization. However, the computational experiments in [6] showed that approximate solutions obtained in this manner do not converge to the exact solutions as the temporal and spatial grid sizes tend to zero. Convergent approximations were then obtained by introducing Tychonoff regularization [6]. In [9], mesh regularization (a two-grid algorithm) or high-order basis functions were also introduced as means for obtaining convergent approximations.

The numerical approximation of the controllability problem (1) was also studied in [10–13]. In these papers, the convergence of the boundary controls for finite-difference discretized equations to the control of the continuous one-dimensional wave equation as the mesh size tends to zero was proved under the assumption that the step size in time equals to that in space, i.e., the CFL condition is satisfied with equality. Furthermore, it was shown in these papers that if the step size in space is less than that in time, then there exists a sequence of controls for the discretized equations that do not converge. Methods of regularization that result in convergent approximations were surveyed in [13]; these include filtering, viscous numerical damping, and the use of mixed finite elements.

Underdetermined system approach

Another computational approach was proposed in [14]. One starts by fully discretizing, e.g., by a finite difference, finite element, or spectral method or by a combination of these, the wave equation (1), the initial conditions (2), and the terminal conditions (3). This results in a linear system of algebraic equations $A\vec{U} = \vec{F}$, where $\vec{U}$ is a vector of unknowns, e.g., the values of a finite-difference approximation of u at the grid points, and $\vec{F}$ is a vector determined from the initial data u_0 and u_1 . Note that the boundary condition (4) is ignored in the determination of A and $\vec{F}$.

In general, the linear system $A\vec{U} = \vec{F}$ is *underdetermined* so that, in principle, *any* of its solutions provides an approximation to the solution of (1)–(3). An approximation g^h to a boundary control can then be determined by setting $g^h = B_h(\vec{U})$, where B_h is a suitable approximation to the operator B and $\vec{U}$ is any particular solution of $A\vec{U} = \vec{F}$. In [14], the normal equation $AA^T \vec{V} = \vec{F}$ was proposed as a means for determining a solution $\vec{U} = A^T \vec{V}$ of the underdetermined linear system $A\vec{U} = \vec{F}$. Of course, the solution obtained in this manner is an approximation to the minimum $L^2(Q)$ -norm (not $L^2(\Sigma)$ -norm) solution of (1)–(3). Also, note that the matrix A couples all time and space levels so that any naive solution strategy is bound to be inefficient.

2. OPTIMIZATION-BASED NUMERICAL METHODS

The methods introduced in this paper incorporate a number of refinements into the method introduced in [14]. First, a particular solution of the underdetermined linear system $A\vec{U} = \vec{F}$ is determined by directly solving an optimization problem for which the equations of the linear system act as equality constraints. The functional to be optimized is a discretization of a functional which reflects the desired property one wishes the solution of the underdetermined controllability problem to satisfy. For example, if one desires a solution of the wave equation (1), the initial conditions (2), and the terminal conditions (3) that also is a solution of the minimization problem (5), we will determine an approximation to this solution by solving the discrete optimization problem

$$\min_{\vec{U}} \mathcal{J}_h(\vec{U}), \quad \text{subject to } A\vec{U} = \vec{F}, \quad (17)$$

where $\mathcal{J}_h$ denotes a discretization of $\mathcal{J}$.

Repeating some observations and anticipating others, our computational results for problems in two space dimensions indicate that our approach has the following strengths.

- No boundary condition on the control part of the boundary need be specified during the solution process; as a result, a single solution can be used to define approximate controls of different types, e.g., Dirichlet or Neumann, by simply evaluating, at the control part of the boundary, an appropriate discrete operator applied to the discrete solution.
- Approximations can be obtained to the solutions of the exact controllability problem that satisfy arbitrary optimization properties, not just the minimum $L^2(\Sigma)$ -norm property.
- For smooth minimum $L^2(\Sigma)$ -norm exact solutions for the control, pointwise convergence of the approximate controls is achieved for any ratio of time to space step sizes satisfying the CFL condition, without the need for regularization.
- For the generic case of nonsmooth minimum $L^2(\Sigma)$ -norm exact solutions for the control, convergence in the L^2 sense of the approximate controls is achieved for any ratio of time to space step sizes satisfying the CFL condition without the need for regularization; pointwise convergence can be achieved through a simple regularization of the functional.
- Approximations of minimum $H^1(\Sigma)$ -norm solutions for the control converge pointwise without the need for regularization.

The one-dimensional case was studied in [15] where extensive computational examples were provided supporting the above observations about our methods. Since the wave equation in one

dimension has special properties that do not extend to higher dimensions, e.g., either independent variable may be viewed as a time-like variable, it is important that the above observations about our methods be confirmed, at least through computations, for higher dimensions; for the two-dimensional case, that is the goal of this paper.

At this point it is important to examine the relation between our results and those of references [1,5,6,9–13]; these comparisons can be made only for problem (9) since the previous methods could only be applied to this problem; recall that our methods can treat more general minimization problems including (11) and (12). Unlike the previous approaches, for problem (9), we are able to obtain convergent approximations (in an $L^2(\Sigma)$ sense for generic solutions that merely belong to $L^2(\Sigma)$ and in a pointwise sense for smoother solutions) for the control without the need for regularization. Since our approach and previous approaches all attempt to compute minimum norm controls, the disparity in the performance of the methods needs to be explained. We will return to this issue in Section 5. However, we note here that our results in no way contradict the theoretical results of [10–13] which state that, in one dimension, there exists a sequence of approximate controls that do not converge. Those results do not in any way address the convergence of a specific sequence of approximate controls. In [15], we provided computational evidence of the convergence of the sequence of approximate controls determined by our methods not only for the minimum $L^2(\Sigma)$ -norm controllability problem, but also for the minimum $H^1(\Sigma)$ -norm and the minimum regularized $L^2(\Sigma)$ -norm controllability problems.

For the sake of concreteness and to be able to compare our method to previous approaches, we first focus on the functional $\mathcal{J}_0(\cdot)$ of (8) or (9), i.e., on the minimum $L^2(\Sigma)$ -norm solution of the controllability problem. Subsequently, we will consider functionals $\mathcal{J}_\epsilon(\cdot)$ of the form (10) with $\epsilon = 1$ as well as with $0 < \epsilon < 1$.

Before considering an optimization problem of the type (17), we first examine some properties of the linear system $A\bar{U} = \bar{F}$. For the sake of concreteness and simplicity, we will consider finite-difference discretizations of (1)–(3) posed on a square domain.

2.1. Coupled Space-Time Finite-Difference Discretization

Let $\Omega = (0, 1) \times (0, 1)$. Then, the minimum value of T for which exact controllability holds is $\sqrt{2}$ so that, throughout, we assume that $T \geq \sqrt{2}$. We employ a finite-difference discretization for the wave equation. Let $x_i = i/J$, $y_j = j/J$, $i, j = 0, 1, \dots, J$, and $t_k = kT/K$, $k = 0, 1, \dots, K$, where J and K are chosen positive integers, denote a partition of $Q = \Omega \times (0, T)$, and let $u_{i,j,k}$ denote the approximation to $u(x_i, y_j, t_k)$. The grid sizes are then given by $\Delta x = \Delta y = h = 1/J$ and $\delta = T/K$. The wave equation (1) is discretized using standard three-point difference quotients [16] so that the $u_{i,j,k}$ s are required to satisfy the following discrete wave equation:

$$\frac{u_{i,j,k+1} - 2u_{i,j,k} + u_{i,j,k-1}}{\delta^2} - \frac{u_{i+1,j,k} + u_{i-1,j,k} + u_{i,j+1,k} + u_{i,j-1,k} - 4u_{i,j,k}}{h^2} = 0, \quad (18)$$

for $i, j = 1, 2, \dots, J-1$ and $k = 1, 2, \dots, K-1$.

Let $\lambda = \delta/h$; then (18) can be simplified to

$$u_{i,j,k+1} - (2 - 4\lambda^2) u_{i,j,k} - \lambda^2 (u_{i+1,j,k} + u_{i-1,j,k} + u_{i,j+1,k} + u_{i,j-1,k}) + u_{i,j,k-1} = 0, \quad (19)$$

for $i, j = 1, 2, \dots, J-1$ and $k = 1, 2, \dots, K-1$.

The standard CFL condition that guarantees the stability of scheme (19) is given by $\lambda \leq 1/\sqrt{2}$ which implies that $TJ/K \leq 1/\sqrt{2}$. Then, since $T \geq \sqrt{2}$, we have

$$K \geq 2J. \quad (20)$$

The initial and terminal conditions are discretized by

$$\begin{aligned} u_{i,j,0} = u_0(x_i, y_j) \quad \text{and} \quad \frac{u_{i,j,1} - u_{i,j,0}}{\delta} &= u_1(x_i, y_j), & \text{for } i, j = 0, 1, \dots, J, \\ u_{i,j,K} = 0 \quad \text{and} \quad \frac{u_{i,j,K} - u_{i,j,K-1}}{\delta} &= 0, & \text{for } i, j = 0, 1, \dots, J. \end{aligned} \quad (21)$$

The linear system formed by (19) and (21) has $p = (J - 1)^2(K - 1) + 4(J + 1)^2$ equations and $q = (J + 1)^2(K + 1)$ unknowns. We symbolically express it in the form

$$A\vec{U} = \vec{F}, \quad (22)$$

where $A \in \mathbb{R}^{p \times q}$ is a banded matrix with bandwidth proportional to $(J + 1)^2$, $\vec{U} \in \mathbb{R}^q$ is a vector of unknowns $u_{i,j,k}$ s, and $\vec{F} \in \mathbb{R}^p$ is a known vector determined from the initial data. It is also worth noting that the number of nonzero elements of each row or each column of A is less than or equal to 7. Furthermore, we have that

$$q - p = (J + 1)^2(K - 3) - (J - 1)^2(K - 1) = 4JK - 2J^2 - 8J - 2 = 4JK - 2J^2 - 8J - 2.$$

From (20), if we assume $J > 1$, then

$$q - p = 4JK - 2J^2 - 8J - 2 \geq 8J^2 - 2J^2 - 8J - 2 = 6J^2 - 8J - 2 \geq 6,$$

i.e., linear system (22) has more unknowns than equations.

Some of the unknowns are uniquely determined by the initial and terminal conditions. First, it is easy to see that all the unknowns appearing in the discrete initial and terminal conditions, i.e., in (21), are uniquely determined

$$\begin{aligned} u_{i,j,0} &= u_0(x_i, y_j) \quad \text{and} \quad u_{i,j,1} = u_0(x_i, y_j) + \delta u_1(x_i, y_j), & \text{for } i, j = 0, 1, \dots, J, \\ u_{i,j,K} &= 0 \quad \text{and} \quad u_{i,j,K-1} = 0, & \text{for } i, j = 0, 1, \dots, J. \end{aligned} \quad (23)$$

Without loss of generality, we assume that $J = 2M + 1$ for some integer $M > 0$. (The case of J being an even integer can be treated in a similar manner.) Then, after computing (23), some of the equations in (19) allow us to uniquely determine some additional unknowns through an explicit marching procedure

$$\begin{aligned} &\text{for } k = 1, \dots, M \\ &\quad k_1 = k, \quad k_2 = K - k \\ &\quad \text{for } i, j = k, \dots, J - k \\ &\quad \quad u_{i,j,k_1+1} = (2 - 4\lambda^2) u_{i,j,k_1} + \lambda^2 (u_{i+1,j,k_1} + u_{i-1,j,k_1} + u_{i,j+1,k_1} + u_{i,j-1,k_1}) - u_{i,j,k_1-1} \\ &\quad \quad u_{i,j,k_2-1} = (2 - 4\lambda^2) u_{i,j,k_2} + \lambda^2 (u_{i+1,j,k_2} + u_{i-1,j,k_2} + u_{i,j+1,k_2} + u_{i,j-1,k_2}) - u_{i,j,k_2+1} \\ &\quad \text{end} \\ &\text{end.} \end{aligned} \quad (24)$$

The unknowns determined by (24) are those associated with the (numerical) domains of dependence of the initial and terminal data and we can reduce the size of system (22) by determining them beforehand. Let $\vec{V}$ be the vector of unknowns determined by (23) and (24); it is easily determined that $\vec{V} \in \mathbb{R}^r$, where $r = 4(J + 1)^2 + J(J^2 - 1)/3$ is also the number of equations in (23) and (24). Let

$$\tilde{p} = p - r = (J - 1)^2(K - 1) - \frac{1}{3}J(J^2 - 1) \quad \text{and} \quad \tilde{q} = q - r = (J + 1)^2(K - 3) - \frac{1}{3}J(J^2 - 1).$$

Clearly, $q - p = \tilde{q} - \tilde{p}$. Let $\tilde{\vec{U}} \in \mathbb{R}^{\tilde{q}}$ be the vector of $u_{i,j,k}$ s not included in $\vec{V}$. Then, the linear system (22) can be reduced to

$$\tilde{A}\tilde{\vec{U}} = \tilde{\vec{F}}, \quad (25)$$

where $\tilde{A} \in \mathbb{R}^{\tilde{p} \times \tilde{q}}$ and $\tilde{\vec{F}} \in \mathbb{R}^{\tilde{q}}$ are obtained by deleting the equations used in (23) and (24) to predetermine $\vec{V}$ and then, in the remaining equations, moving the terms involving the components of $\vec{V}$ to the right-hand side of (22).

Regarding linear system (25), we have the following result.

PROPOSITION 2.1. $\text{rank}(\tilde{A}) = \tilde{p}$, i.e., the equations included in system (25) are linearly independent.

PROOF. Recall that $J = 2M + 1$. Let

$$\begin{aligned}\bar{W}_0 &= \{u_{i,j,k} \mid k = M + 2, \dots, K - M - 2, i, j = M, M + 1\}, \\ \bar{W}_1 &= \{u_{i,j,k} \mid j = 0, \dots, M - 1, i = j, j + 1, k = j + 2, \dots, K - j - 2\}, \\ \bar{W}_2 &= \{u_{i,j,k} \mid j = 0, \dots, M - 1, i = J - j - 1, J - j, k = j + 2, \dots, K - j - 2\}, \\ \bar{W}_3 &= \{u_{i,j,k} \mid j = M + 2, \dots, J, i = j - 1, j, k = J - j + 2, \dots, K - J + j - 2\}, \\ \bar{W}_4 &= \{u_{i,j,k} \mid j = M + 2, \dots, J, i = J - j, J - j + 1, k = J - j + 2, \dots, K - J + j - 2\}.\end{aligned}\quad (26)$$

See Figure 1 for a visual depiction of $\bar{W}_i$. Then, let $\vec{W} \subset \vec{U}$ be the vector of all unknowns in $\bar{W}_i$ for $i = 0, 1, \dots, 4$; one easily finds that $\vec{W} \in \mathbb{R}^s$, where

$$s = 4(K - 2M - 3) + 8(K - M - 2)M = 8MK + 4K - 8M^2 - 24M - 12.$$

It is not difficult to verify that, with $J = 2M + 1$, $s = q - p = \tilde{q} - \tilde{p}$. Thus, the cardinality of $\vec{W}$ is exactly the difference between the number of columns and rows of A or $\tilde{A}$. (Since $\tilde{q} > \tilde{p}$, equation (25) is an *underdetermined* linear system.)

Assume that $\vec{W}$ is known. Since $\vec{V}$ has already been determined and is involved in $\vec{F}$, the right-hand side of (25), and

$$K \geq 2J = 2(2M + 1) = 4M + 2 \geq 2(M + 2), \quad \text{for } M > 0,$$

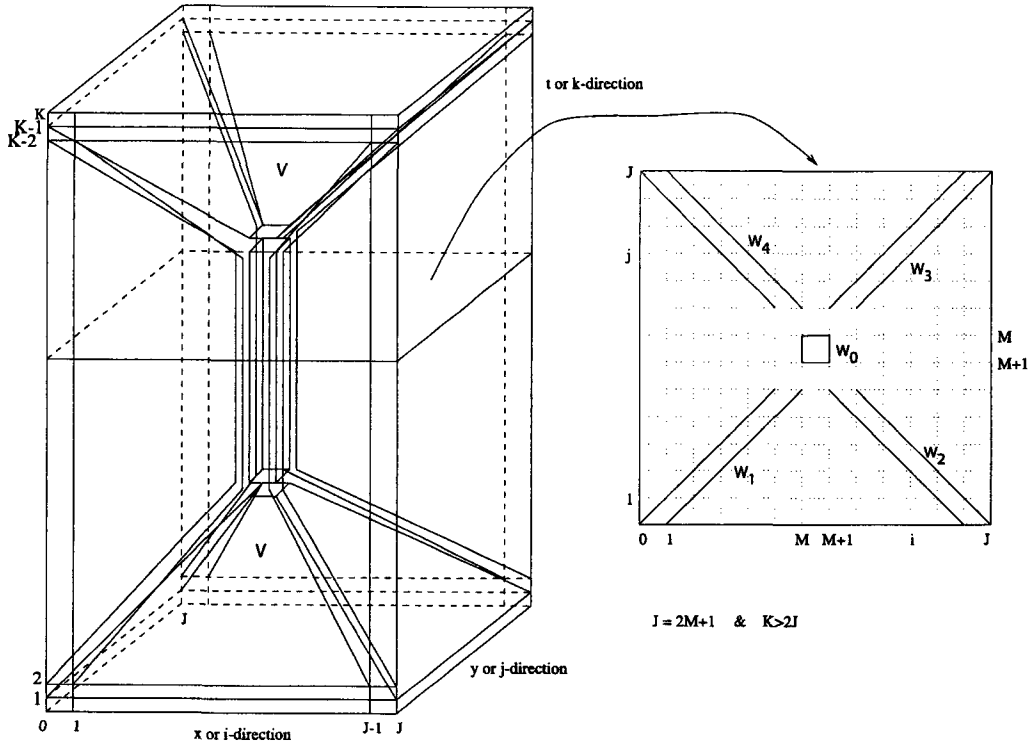


Figure 1. Visualization of $\vec{W}$.

we can re-express the linear system (25) by the following recursive computations, i.e., by a marching procedure from the center of Q to the lateral surface Σ :

$$\begin{aligned}
 &\text{for } j = M - 1, \dots, 1 \\
 &\quad j_1 = j, \quad j_2 = J - j \\
 &\quad \text{for } i = j + 1, \dots, J - j - 1 \\
 &\quad \quad \text{for } k = j + 1, \dots, K - j - 1 \\
 &\quad \quad \quad u_{i,j_1-1,k} = \frac{1}{\lambda^2} (u_{i,j_1,k+1} + u_{i,j_1,k-1}) - \left(\frac{2}{\lambda^2} - 4 \right) u_{i,j_1,k} \\
 &\quad \quad \quad \quad - (u_{i+1,j_1,k} + u_{i-1,j_1,k} + u_{i,j_1+1,k}) \\
 &\quad \quad \quad u_{i,j_2+1,k} = \frac{1}{\lambda^2} (u_{i,j_2,k+1} + u_{i,j_2,k-1}) - \left(\frac{2}{\lambda^2} - 4 \right) u_{i,j_2,k} \\
 &\quad \quad \quad \quad - (u_{i+1,j_2,k} + u_{i-1,j_2,k} + u_{i,j_2-1,k}) \\
 &\quad \quad \text{end} \\
 &\quad \text{end} \\
 &\text{end}
 \end{aligned} \tag{27}$$

and

$$\begin{aligned}
 &\text{for } i = M, \dots, 1 \\
 &\quad i_1 = i, \quad i_2 = J - i \\
 &\quad \text{for } j = i, \dots, J - i \\
 &\quad \quad \text{for } k = j + 1, \dots, K - j - 1 \\
 &\quad \quad \quad u_{i_1-1,j,k} = \frac{1}{\lambda^2} (u_{i_1,j,k+1} + u_{i_1,j,k-1}) - \left(\frac{2}{\lambda^2} - 4 \right) u_{i_1,j,k} \\
 &\quad \quad \quad \quad - (u_{i_1,j-1,k} + u_{i_1,j+1,k} + u_{i_1+1,j,k}) \\
 &\quad \quad \quad u_{i_2+1,j,k} = \frac{1}{\lambda^2} (u_{i_2,j,k+1} + u_{i_2,j,k-1}) - \left(\frac{2}{\lambda^2} - 4 \right) u_{i_2,j,k} \\
 &\quad \quad \quad \quad - (u_{i_2,j-1,k} + u_{i_2,j+1,k} + u_{i_2-1,j,k}) \\
 &\quad \quad \text{end} \\
 &\quad \text{end} \\
 &\text{end.}
 \end{aligned} \tag{28}$$

By the above procedure, we finally have determined all the components of $\tilde{U}$ in the linear system (25) and we have thus shown that $\text{rank}(\tilde{A}) = \tilde{q} - s = \tilde{p}$, i.e., by setting $\tilde{W}$, i.e., s of the components in $\tilde{U}$, equation (25) uniquely determines the remaining components. Thus, $\tilde{A}$ has full row rank and the equations included in (25) are linearly independent.

Of course, Proposition 2.1 implies that coefficient matrix in (22) is also of full row rank. Also, the proposition implies that, once $\tilde{V}$ is determined from the initial and terminal date, if $\tilde{W}$ is specified, then the remainder of $\tilde{U}$ can be determined from the marching procedure (27),(28).

2.2. Constrained Quadratic Optimization System

We now turn to the selection of a particular solution of the linear system (22), i.e., of the discretized wave equation and initial and terminal conditions, by requiring that solution to also solve a discrete optimization problem; see (17). For the time being, we focus on a discretization of the optimization problem (9) to seek an approximation of the minimum $L^2(\Sigma)$ -norm solution of the controllability problem. In particular, we discretize the functional $\mathcal{J}_0(\cdot)$ of that proposition

using the trapezoidal rule to obtain the approximation

$$\begin{aligned} \mathcal{J}_0(u) &\equiv \int_{\Sigma} u^2 d\Gamma dt = \int_0^T \int_0^1 (u^2(z, 0, t) + u^2(z, 1, t) + u^2(0, z, t) + u^2(1, z, t)) dz dt \\ &\approx h\delta \sum_{k=0}^K \alpha_k \left(\sum_{i=0}^J (u_{i,0,k}^2 + u_{i,J,k}^2) + \sum_{j=1}^{J-1} (u_{0,j,k}^2 + u_{J,j,k}^2) \right) \equiv \mathcal{J}_{0,h}(\tilde{U}), \end{aligned} \quad (29)$$

where $\alpha_0 = \alpha_K = 1/2$ and $\alpha_1 = \dots = \alpha_{K-1} = 1$. Furthermore, we effect the partition $\tilde{U} = (\tilde{U}_1^\top, \tilde{U}_2^\top)^\top$, where $\tilde{U}_1$ denotes the subvector of $\tilde{U}$ belonging to the lateral surface $\tilde{\Sigma}$ and $\tilde{U}_2$ denotes the subvector of $\tilde{U}$ belonging to the interior Q . It is seen that $\tilde{U}_1 \in \mathbb{R}^{\tilde{q}_1}$, where $\tilde{q}_1 = 4J(K-3)$, and $\tilde{U}_2 \in \mathbb{R}^{\tilde{q}_2}$, where $\tilde{q}_2 = \tilde{q} - 4J(K-3)$. We are then led to the following *discrete* quadratic optimization problem corresponding to (9):

$$\min_{\tilde{U} \in \mathbb{R}^{\tilde{q}}} \mathcal{J}_{0,h}(\tilde{U}), \quad \text{subject to } \tilde{A}_1 \tilde{U}_1 + \tilde{A}_2 \tilde{U}_2 = \tilde{F}, \quad (30)$$

where $\mathcal{J}_{0,h}(\tilde{U}) = h\delta(\tilde{U}_1^\top \tilde{U}_1 + b)$, $\tilde{A}_1 \in \mathbb{R}^{\tilde{p} \times \tilde{q}_1}$, and $\tilde{A}_2 \in \mathbb{R}^{\tilde{p} \times \tilde{q}_2}$ are obtained by appropriately rearranging and partitioning $\tilde{A}$ corresponding to $\tilde{U}_1$ and $\tilde{U}_2$, and b is a constant depending on the components of $\tilde{U}$ belonging to $\tilde{\Sigma}$ that are predetermined by (23). It should be pointed out that problem (30) couples space and time together.

Regarding the solution of the optimization system (30), we have the following result.

PROPOSITION 2.2. *The discrete quadratic optimization system (30) has a unique solution.*

PROOF. Existence is trivial since $\mathcal{J}_{0,h}(\tilde{U})$ is a convex functional of $\tilde{U}$ and, by Proposition 2.1, $\tilde{A}\tilde{U} = \tilde{F}$ is a set of linear, independent constraints.

Now, suppose that both $\tilde{U} = (\tilde{U}_1, \tilde{U}_2)$ and $\tilde{Z} = (\tilde{Z}_1, \tilde{Z}_2)$ are solutions of the optimization problem (30). Then,

$$\mathcal{J}_{0,h}(\tilde{U}) = h\delta \left(\tilde{U}_1^\top \tilde{U}_1 + b \right) = h\delta \left(\tilde{Z}_1^\top \tilde{Z}_1 + b \right) = \mathcal{J}_{0,h}(\tilde{Z}). \quad (31)$$

If we define $\tilde{\mathcal{J}}_{0,h}(\tilde{U}_1) = h\delta(\tilde{U}_1^\top \tilde{U}_1 + b)$ for $\tilde{U}_1 \in \mathbb{R}^{\tilde{q}_1}$, it is easy to show that $\tilde{\mathcal{J}}_{0,h}$ is a strictly convex function of $\tilde{U}_1$. Then, the unconstrained minimization problem of minimizing $\tilde{\mathcal{J}}_{0,h}(\tilde{U}_1)$ over $\tilde{U}_1 \in \mathbb{R}^{\tilde{q}_1}$ has a unique solution and, from (31), we conclude that $\tilde{U}_1 = \tilde{Z}_1$. With the same discrete initial conditions (23), we then obtain that $\tilde{U}_2 = \tilde{Z}_2$ by explicit marching. Thus, $\tilde{U} = \tilde{Z}$.

2.3. Solution Processes and Implementation

We now discuss how the discrete optimization problem (30) can be solved.

A Lagrange multiplier method

If $\tilde{U}$ is the solution of the optimization problem (30), then the Lagrange multiplier rule states that $\tilde{U}$ is a stationary point of the Lagrangian functional

$$\mathcal{L}(\tilde{U}) = h\delta \left(\tilde{U}_1^\top \tilde{U}_1 + b \right) - \tilde{S}^\top \left(\tilde{A}_1 \tilde{U}_1 + \tilde{A}_2 \tilde{U}_2 - \tilde{F} \right),$$

where $\tilde{S} \in \mathbb{R}^{\tilde{p}}$ is the Lagrange multiplier introduced to enforce the constraint equations (25). Then, it is well known that $\tilde{U} = (\tilde{U}_1, \tilde{U}_2)$ and $\tilde{S}$ are the solution of the system

$$\begin{pmatrix} I_{\tilde{q}_1} & 0 & \tilde{A}_1^\top \\ 0 & 0 & \tilde{A}_2^\top \\ \tilde{A}_1 & \tilde{A}_2 & 0 \end{pmatrix} \begin{pmatrix} \tilde{U}_1 \\ \tilde{U}_2 \\ \tilde{S} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \tilde{\bar{F}} \end{pmatrix}, \quad (32)$$

where $\tilde{\bar{F}} = \tilde{S}/(h\delta)$. System (32) is easily seen to be equivalent to

$$\begin{pmatrix} 0 & \tilde{A}_2^\top \\ \tilde{A}_2 & -\tilde{A}_1^\top \tilde{A}_1 \end{pmatrix} \begin{pmatrix} \tilde{U}_2 \\ \tilde{S} \end{pmatrix} = \begin{pmatrix} 0 \\ \tilde{\bar{F}} \end{pmatrix} \quad (33)$$

and

$$\tilde{U}_1 = -\tilde{A}_1^\top \tilde{S}. \quad (34)$$

Thus, we need to first solve the linear system (33) and then use (34) to determine $\tilde{U}_1$. Note that $\tilde{U}_1$ along with the boundary $u_{i,j,k}$ s found from (23) and (24) are the approximation to the exact boundary control $u|_\Sigma = g$ of (9) we are seeking.

Clearly, the coefficient matrix of the linear system (33) is symmetric but it is not positive definite. Note that $\tilde{A}_1$ and $\tilde{A}_2$ are sparse matrices having, like A , at most seven nonzero elements in each of their rows or columns. Thus, it is efficient to use iterative methods, e.g., BiCG, SYM-MQL, or CGNR, to solve (33); the application of a preconditioner is useful since the dimension of system (33) is large. In a practical implementation, the matrices $\tilde{A}_1$ and $\tilde{A}_2$ need not be stored. It is also worth noting that, at each step of an iteration, we may only need to update $\tilde{S}$ and that part of $\tilde{W}$ included in $\tilde{U}_2$; then the remainder of $\tilde{U}_2$ can be directly computed by a marching procedure.

Reduction to an unconstrained optimization problem

We next consider another possible efficient method to solve the optimization problem (30). Using the marching procedure (27) and (28), we can find all the unknowns $\tilde{U}_1$ from $\tilde{W}$, provided that $\tilde{V}$ is precalculated using (23) and (24). Thus, $\tilde{U}_1$ is a linear function of $\tilde{W}$, i.e., there exists a matrix $D \in \mathbb{R}^{\tilde{q}_1 \times s}$ such that

$$\tilde{U}_1 = D\tilde{W} + \tilde{C}, \quad (35)$$

where $\tilde{C} \in \mathbb{R}^{\tilde{q}_1}$ is determined from $\tilde{V}$. Then, the optimization problem (30) is equivalent to

$$\min_{\tilde{W} \in \mathbb{R}^r} \mathcal{J}_{0,h}(\tilde{U}_1), \quad \text{subject to } \tilde{U}_1 = D\tilde{W} + \tilde{C}. \quad (36)$$

Compared to (30), we now only need to solve a two-dimensional optimization problem.

Substituting (35) into $\mathcal{J}_{0,h}(\tilde{U}_1)$, we obtain

$$\mathcal{J}_{0,h}(\tilde{U}_1) = (D\tilde{W} + \tilde{C})^\top (D\tilde{W} + \tilde{C}) = \tilde{W}^\top (D^\top D) \tilde{W} + \tilde{C}^\top D\tilde{W} + \tilde{W}^\top D^\top \tilde{C} + \tilde{C}^\top \tilde{C}$$

so that the constrained optimization problem (36) is equivalent to the *unconstrained* optimization problem

$$\min_{\tilde{W} \in \mathbb{R}^r} \hat{\mathcal{J}}_{0,h}(\tilde{W}), \quad \text{where } \hat{\mathcal{J}}_{0,h}(\tilde{W}) \equiv \tilde{W}^\top (D^\top D) \tilde{W} + 2\tilde{C}^\top D\tilde{W} + \tilde{C}^\top \tilde{C}. \quad (37)$$

Since $\vec{W}$ is a stationary point of $\hat{\mathcal{J}}_{0,h}(\vec{W})$, we obtain

$$D^\top D \vec{W} = -2D^\top \vec{C}. \quad (38)$$

Since $D^\top D$ is a positive definite matrix, we can use, e.g., the conjugate gradient method to solve the linear system (38).

The matrix $D \in \mathbb{R}^{\tilde{q}_1 \times s}$ is a full matrix and $\tilde{q}_1$ is a large number even for relatively small J and K so that directly storing D requires large amounts of memory. However, recall that D embodies the marching procedure (27) and (28); that procedure is done level by level from the center of the space-time cylinder Q to its lateral surface Σ . In fact, the marching procedure can be expressed as

$$\begin{aligned} \vec{U}_{M-1} &= D_{3,M-1} \vec{W} + \vec{C}_{M-1}, \\ \vec{U}_{M-2} &= D_{2,M-2} \vec{U}_{M-1} + D_{3,M-2} \vec{W} + \vec{C}_{M-2}, \\ \vec{U}_m &= D_{1,m} \vec{U}_{m+1} + D_{2,m} \vec{U}_{m+2} + D_{3,m} \vec{W} + \vec{C}_m, \quad \text{for } m = M-3, \dots, 0, \end{aligned}$$

where

$$\vec{U}_m = \left\{ u_{i,j,k} \in \vec{U} \mid i = m \text{ or } i = J - m \text{ or } j = m \text{ or } j = J - m \right\}, \quad \text{for } m = 0, 1, \dots, M-1,$$

and the $\{D_{i,m}\}$ are matrices. Clearly, $\vec{U}_1 = \vec{U}_0$ and each $D_{i,m}$ is a banded matrix having at most seven nonzero elements in each of its rows or columns and D can be represented by the sequence of matrices $\{D_{i,m}\}$ corresponding to each level of the marching procedure (27) and (28). Consequently, we need only save $\{D_{i,m}\}$ to completely determine the action of D and $D^\top$ on a vector.

Note that this implementation cannot be directly applied to implicit (in time) approximations of the wave equation or to finite-element approximations involving (unlumped) mass matrices.

3. COMPUTATIONAL EXPERIMENTS FOR MINIMUM $L^2(\Sigma)$ -NORM BOUNDARY CONTROLS

3.1. The Example of Glowinski, Li, and Lions

We first consider the remarkable example given in [6]. Suppose that $\Omega = (0, 1) \times (0, 1)$ and $T = 15/(4\sqrt{2})$. Let $\mathbf{x} = (x, y)$ and set

$$\psi_0(\mathbf{x}, t) = -\sqrt{2}\pi \cos \sqrt{2}\pi \left(t - \frac{1}{4\sqrt{2}} \right) (\sin \pi x \cos 2\pi y + \cos 2\pi x \sin \pi y)$$

and

$$\begin{aligned} \psi_1(\mathbf{x}, t) &= \left[4\pi(T-t) \sin \sqrt{2}\pi \left(t - \frac{1}{4\sqrt{2}} \right) - \frac{28}{2\sqrt{2}} \sin \sqrt{2}\pi(t-T) \right] \sin \pi x \sin \pi y \\ &\quad + 4 \sin \pi x \sum_{p \geq 3 \text{ and } p \text{ odd}} \frac{p}{p^2-1} \left[\frac{2}{\sqrt{1+p^2}} \sin \pi \sqrt{1+p^2}(t-T) \right. \\ &\quad \left. + \frac{3\sqrt{2}}{p^2-4} \cos \sqrt{2}\pi \left(t - \frac{1}{4\sqrt{2}} \right) \right] \sin p\pi y \\ &\quad + 4 \sin \pi y \sum_{p \geq 3 \text{ and } p \text{ odd}} \frac{p}{p^2-1} \left[\frac{2}{\sqrt{1+p^2}} \sin \pi \sqrt{1+p^2}(t-T) \right. \\ &\quad \left. + \frac{3\sqrt{2}}{p^2-4} \cos \sqrt{2}\pi \left(t - \frac{1}{4\sqrt{2}} \right) \right] \sin p\pi x. \end{aligned}$$

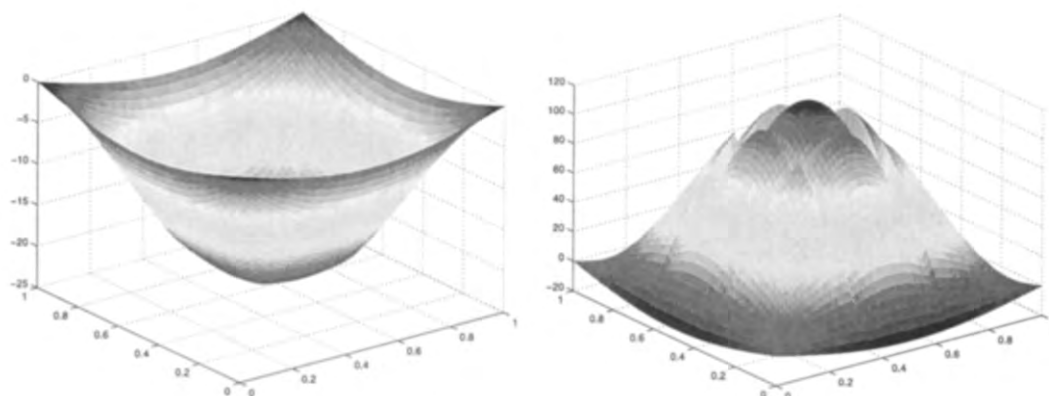


Figure 2. The initial data $u_0(\mathbf{x})$ (left) and $u_1(\mathbf{x})$ (right) for the example of [6].

Then, set

$$u_0(\mathbf{x}) = \psi_0(\mathbf{x}, 0) + \psi_1(\mathbf{x}, 0) \quad \text{and} \quad u_1(\mathbf{x}) = \frac{\partial \psi_0}{\partial t}(\mathbf{x}, 0) + \frac{\partial \psi_1}{\partial t}(\mathbf{x}, 0). \quad (39)$$

Figure 2 provides plots of u_0 and u_1 . Note that u_0 is a Lipschitz continuous function but does not belong to $C^1(\bar{\Omega})$ and u_1 is a bounded function but does not belong to $C^0(\bar{\Omega})$.

Now, let

$$e_0(\mathbf{x}) = \sin \pi x \sin \pi y \quad \text{and} \quad e_1(\mathbf{x}) = \sqrt{2}\pi \sin \pi x \sin \pi y.$$

It is shown in [6] that $\Lambda(e_0, e_1) = (u_1, -u_0)$, so that

$$\psi(\mathbf{x}, t) = \psi_0(\mathbf{x}, t) + \psi_1(\mathbf{x}, t) \quad (40)$$

is the exact solution of optimization problem (9). The corresponding exact control g is then given by restricting $\psi(\mathbf{x}, t)$ to the lateral surface Σ , i.e.,

$$\begin{aligned} g|_{\Sigma_1 \cup \Sigma_2} &= -\sqrt{2}\pi \cos \sqrt{2}\pi \left(t - \frac{1}{4\sqrt{2}} \right) \sin \pi y, \\ g|_{\Sigma_3 \cup \Sigma_4} &= -\sqrt{2}\pi \cos \sqrt{2}\pi \left(t - \frac{1}{4\sqrt{2}} \right) \sin \pi x, \end{aligned} \quad (41)$$

where

$$\begin{aligned} \Sigma_1 &= \{\mathbf{x} \in \Sigma \mid x = 0\}, & \Sigma_2 &= \{\mathbf{x} \in \Sigma \mid x = 1\}, \\ \Sigma_3 &= \{\mathbf{x} \in \Sigma \mid y = 0\}, & \text{and} \quad \Sigma_4 &= \{\mathbf{x} \in \Sigma \mid y = 1\}. \end{aligned}$$

To summarize, equation (41) gives the minimum $L^2(\Sigma)$ -norm boundary control g that solves the exact controllability problem (1)–(3) for the initial data (39); the corresponding exact solution u in Q is given by (40), i.e., $u(\mathbf{x}, t) = \psi(\mathbf{x}, t)$. For future reference, note that $\|g\|_{L^2(\Sigma)} \approx 7.3866$ and $\|u\|_{L^2(Q)} \approx 10.7932$.

We apply the numerical method discussed in Section 2 to the example of [6]. The initial data u_0 and u_1 are given by (39). Computational experiments were carried out for $h = 1/16, 1/32, 1/64$, and $1/128$ with $\delta = 15h/(16\sqrt{2})$ so that $(J, K) = (16, 64), (32, 128), (64, 256)$, and $(128, 512)$, respectively, and the stability condition (20) is satisfied.

The results of our numerical experiments are summarized in Table 1, where u^h and g^h are the computed approximations of u and g , respectively. All norms were calculated by linearly interpolating the discrete values of u^h and g^h . From this table, it seems that g^h converges to g in

Table 1. Results of the computational experiments for the example of [6].

h	$\frac{1}{16}$	$\frac{1}{32}$	$\frac{1}{64}$	$\frac{1}{128}$
$\ g^h\ _{L^2(\Sigma)}$	7.1198	7.3011	7.3291	7.3491
$\frac{\ g^h - g\ _{L^2(\Sigma)}}{\ g\ _{L^2(\Sigma)}}$	0.08917	0.03875	0.01539	0.00699
$\ u^h\ _{L^2(Q)}$	10.3861	10.6672	10.7166	10.7448
$\frac{\ u^h - u\ _{L^2(Q)}}{\ u\ _{L^2(Q)}}$	0.07168	0.02584	0.01416	0.00792

the $L^2(\Sigma)$ -norm at a rate a little higher than first order. The results for $h = 1/128$ are especially noteworthy because at that value of h , the unregularized method of [6] clearly displays unstable and nonconvergent behavior; in contrast, no such aberrant behavior occurs for our method for that value of h .

In order to visualize the convergence of our method as h becomes smaller, in Figures 3–6 we provide plots of the exact control g and the corresponding computed discrete control g^h for several values of h . Comparisons over time for several fixed points on the boundary and over a side of the boundary for fixed values of time are provided. In Figure 7, we also compare the function $t \rightarrow \|g(t)\|_{L^2(\Gamma)}$ with $t \rightarrow \|g^h(t)\|_{L^2(\Gamma)}$. To complete this presentation of the computational results related to the example of [6], a visualization of the evolution of the system to the zero terminal data is given in Figure 8; more precisely, we provide, for $h = 1/128$, plots of $u^h(\cdot, \cdot, t)$ at various instants of time. From these figures, it seems that for the example of [6], our method produces pointwise convergent approximations *without the need for regularization*. Additionally, we remark

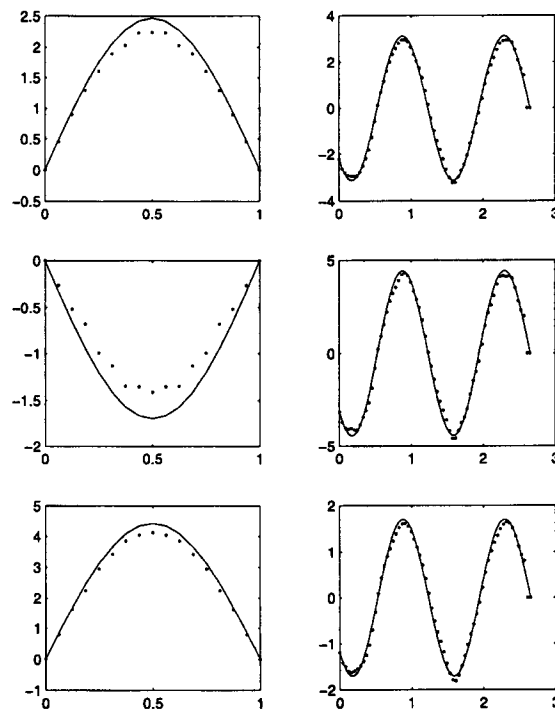


Figure 3. The exact control g (—) and the approximate control g^h (···) for the example of [6] for $h = 1/16$ evaluated at $(0, y, T/4)$ (left-top), $(0, y, T/2)$ (left-middle), $(0, y, T/8)$ (left-bottom), $(1/4, 0, t)$ (right-top), $(1/2, 0, t)$ (right-middle), and $(7/8, 0, t)$ (right-bottom).

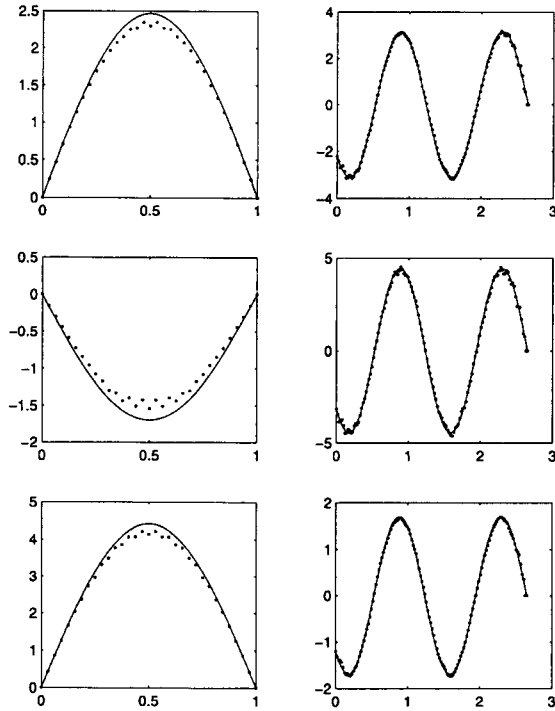


Figure 4. The exact control g (—) and the approximate control g^h (···) for the example of [6] for $h = 1/32$ evaluated at $(0, y, T/4)$ (left-top), $(0, y, T/2)$ (left-middle), $(0, y, 7T/8)$ (left-bottom), $(1/4, 0, t)$ (right-top), $(1/2, 0, t)$ (right-middle), and $(7/8, 0, t)$ (right-bottom).

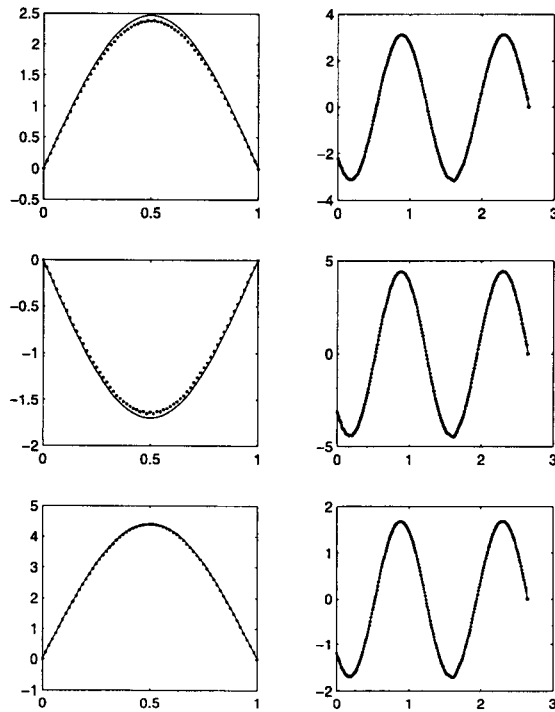


Figure 5. The exact control g (—) and the approximate control g^h (···) for the example of [6] for $h = 1/64$ evaluated at $(0, y, T/4)$ (left-top), $(0, y, T/2)$ (left-middle), $(0, y, 7T/8)$ (left-bottom), $(1/4, 0, t)$ (right-top), $(1/2, 0, t)$ (right-middle), and $(7/8, 0, t)$ (right-bottom).

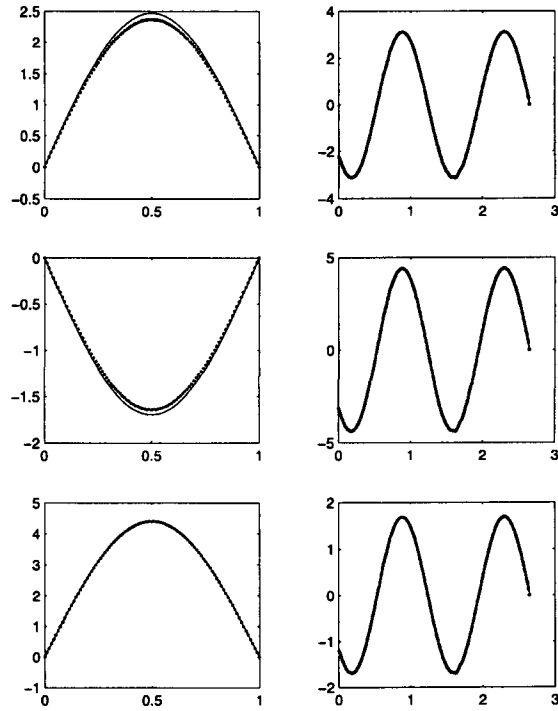


Figure 6. The exact control g (—) and the approximate control g^h (···) for the example of [6] for $h = 1/128$ evaluated at $(0, y, T/4)$ (left-top), $(0, y, T/2)$ (left-middle), $(0, y, 7T/8)$ (left-bottom), $(1/4, 0, t)$ (right-top), $(1/2, 0, t)$ (right-middle), and $(7/8, 0, t)$ (right-bottom).

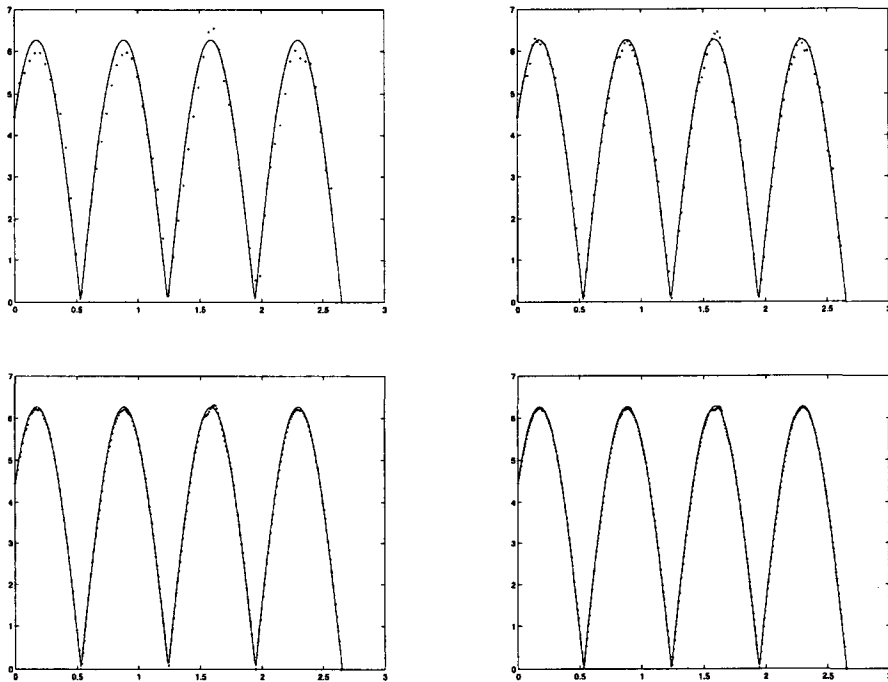


Figure 7. The norms $\|g(t)\|_{L_2(\Gamma)}$ of the exact control (—) and $\|g^h(t)\|_{L_2(\Gamma)}$ of the approximate control (···) for the example of [6] for $h = 1/16$ (top-left), $h = 1/32$ (top-right), $h = 1/64$ (bottom-left), and $h = 1/128$ (bottom-right).

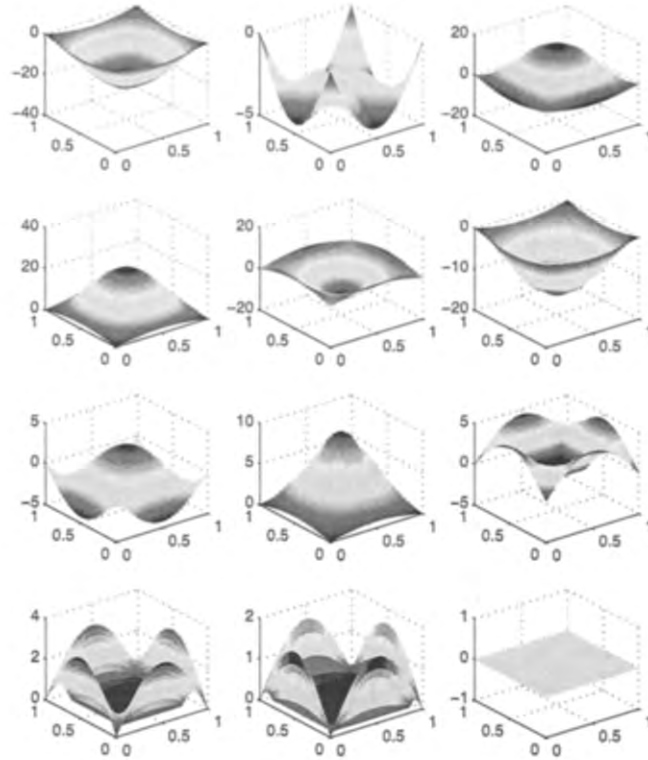


Figure 8. Visualization of the approximate solution $u^h(x, y, t)$ for the example of [6] for $h = 1/128$ at $t = 0, T/16, T/8, T/4, 3T/8, T/2, 5T/8, 3T/4, 7T/8, 15T/16, 31T/32$, and T (from left to right and from top to bottom).

that some other choices of $\lambda = \delta/h$ were also tested and the convergence of approximations was always obtained so long as the CFL condition was satisfied, i.e., $\lambda \leq 1/\sqrt{2}$.

The observation that our method produces convergent approximations for any value of λ that satisfies the CFL condition without the need for regularization at first seems to contradict results of, e.g., [11] that state that, at least in one dimension, for λ strictly less than the CFL number, there exists a sequence of approximate controls that do not converge to the exact control. However, that result says nothing about the *particular* sequence of minimum discrete $\ell^2(\Sigma)$ -norm controls that is determined by our method. Thus, the seeming convergence of our sequence of discrete controls is not contradictory. See Section 5 and [15] for further discussions of this point.

3.2. Generic Examples

In the example of [6] just discussed, the minimum $L^2(\Sigma)$ -norm control is very smooth. Using our methods, we obtained good approximations for this example. However, this is not the generic case. In general, even for smooth initial data, the minimum $L^2(\Sigma)$ -norm control for the controllability problem (1)–(3) will not be smooth. See [15] for some examples in the one-dimensional setting which indicate that generic solutions are piecewise smooth in time, i.e., they contain jump discontinuities. In this section, we further illustrate this point and show that our methods, when applied to the minimum $L^2(\Sigma)$ -norm controllability problem, seemingly converge (in an L^2 sense) even in the generic case.

We consider two examples having smooth, i.e., $C^\infty(\bar{\Omega})$, initial conditions. With $\Omega = (0, 1) \times (0, 1)$ and $T = 2$, we let

$$\begin{aligned} \text{Case I: } u_0(\mathbf{x}) &= \sin(\pi x) \sin(\pi y) \quad \text{and} \quad u_1(\mathbf{x}) = \sqrt{2}\pi \sin(\pi x) \sin(\pi y). \\ \text{Case II: } u_0(\mathbf{x}) &= \sin(x) \sin(y) \quad \text{and} \quad u_1(\mathbf{x}) = \sqrt{2} \sin(x) \sin(y). \end{aligned} \tag{42}$$

The initial conditions for Case I vanish on the boundary Γ ; those for Case II do not.

Table 2. Results of the computational experiments for the minimum $L^2(\Sigma)$ -norm case for the generic examples with initial data (42).

	h	$\frac{1}{16}$	$\frac{1}{32}$	$\frac{1}{64}$
Case I	$\ g^h\ _{L^2(\Sigma)}$	0.4134	0.3939	0.3850
	$\ u^h\ _{L^2(Q)}$	0.5203	0.5078	0.5015
Case II	$\ g^h\ _{L^2(\Sigma)}$	0.2322	0.1985	0.1753
	$\ u^h\ _{L^2(Q)}$	0.1939	0.1860	0.1811

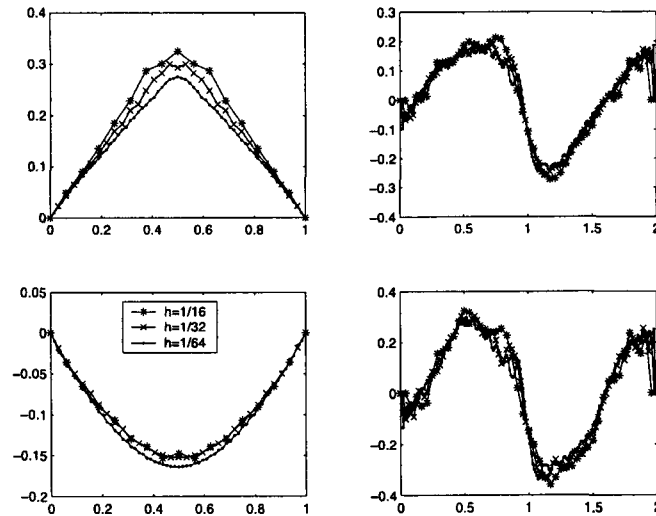


Figure 9. The approximation g^h that has minimum $L^2(\Sigma)$ -norm for the generic example with initial data (42)-I evaluated at $(0, y, T/4)$ (left-top); $(0, y, T/2)$ (left-bottom); $(1/4, 0, t)$ (right-top); and $(1/2, 0, t)$ (right-bottom).

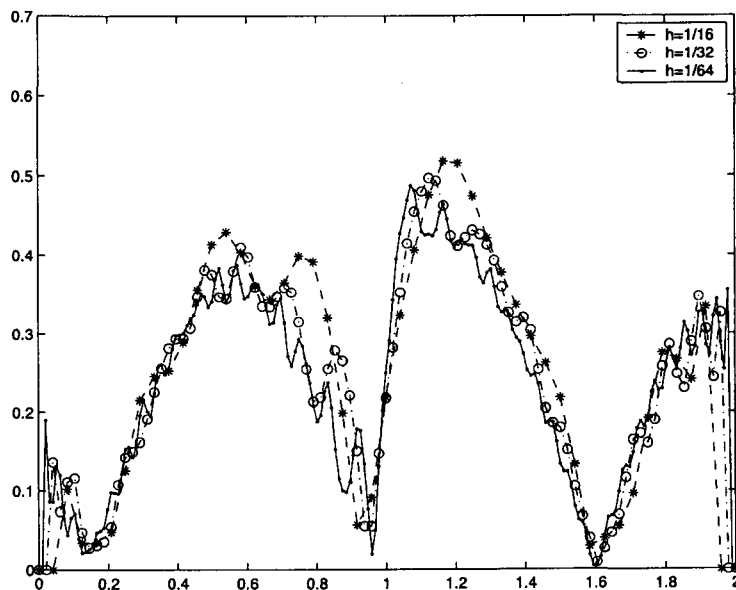


Figure 10. The norm $\|g^h(t)\|_{L^2(\Gamma)}$ of the approximate control that has minimum $L^2(\Sigma)$ -norm for the generic example with initial data (42)-I for $h = 1/16, 1/32$, and $1/64$.

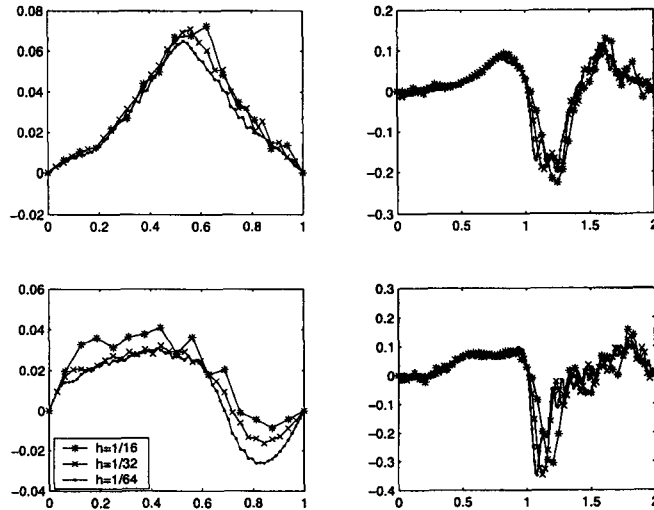


Figure 11. The approximation g^h that has minimum $L^2(\Sigma)$ -norm for the generic example with initial data (42)-II evaluated at $(0, y, T/4)$ (left-top); $(0, y, T/2)$ (left-bottom); $(1/4, 0, t)$ (right-top); and $(1/2, 0, t)$ (right-bottom).

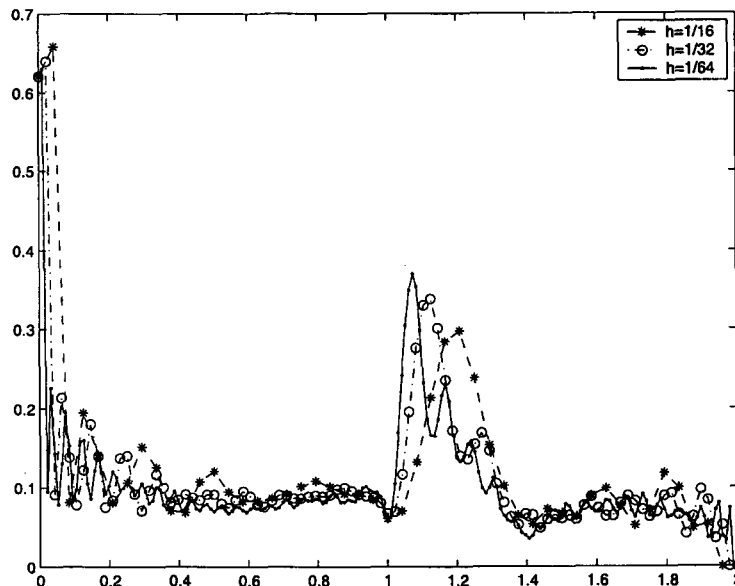


Figure 12. The norm $\|g^h(t)\|_{L^2(\Gamma)}$ of the approximate control that has minimum $L^2(\Sigma)$ -norm for the generic example with initial data (42)-II for $h = 1/16, 1/32$, and $1/64$.

Computational experiments were carried out for the spatial grid sizes $h = 1/16, 1/32$, and $1/64$ along with the corresponding $\delta = 2h/3$ so that $(J, K) = (16, 48), (32, 96)$, and $(64, 192)$, respectively, and the stability condition (20) is satisfied. The results of the computational experiments are summarized in Table 2. Of course, since we do not know the exact solution for this example, we cannot list any information about the exact control function g or the exact state u .

In Figure 9 for Case I and in Figure 11 for Case II, we provide plots of the computed discrete solution g^h for several values of h . Plots over time for several fixed points on the boundary and over a side of the boundary for fixed values of time are provided. In Figure 10 for Case I and Figure 12 for Case II, we show the function $t \rightarrow \|g^h(t)\|_{L^2(\Gamma)}$ for several values of h .

Figures 9–12 clearly show that, as the temporal and spatial grids are refined, the computed approximation to the minimum $L^2(\Sigma)$ -norm control becomes more and more oscillatory in both

space and time. Note, however, that although the frequencies of the oscillations increase as the grid spacing decreases, the amplitudes of the oscillations do not; if anything, the amplitudes seem to decrease with decreasing grid size. Furthermore, Table 2 indicates that the $L^2(\Sigma)$ -norm of the approximate control g^h and the $L^2(Q)$ -norm of the approximate solution u^h converge as h and δ tend to zero. The figures also seemingly show that, at least with respect to time, the minimum $L^2(\Sigma)$ -norm control contains jump discontinuities. These observations are consistent with the one-dimensional case; see [15]. Extrapolating from that case and using the observations gleaned from Table 2 and Figures 9–12 lead us to conclude that:

- the oscillatory behavior of the approximate solution is a numerical artifact, but that the jump discontinuities are not so that, in general, the minimum $L^2(\Sigma)$ -norm solution of the exact controllability problem (1)–(3) is not smooth, i.e., it is not better than $L^2(\Sigma)$; the example of [6] involving a smooth control is thus a very special case that is not representative of the generic situation;
- despite their oscillatory behavior, the approximate solutions obtained using our methods converge in $L^2(\Sigma)$ (for the approximate Dirichlet control g^h) and in $L^2(Q)$ (for the approximate state u^h) even when the minimum $L^2(\Sigma)$ -norm control is not smooth.

We again note that these observations do not contradict the nonconvergence result of, e.g., [11].

One may be able to remove the oscillations from our approximate solutions by an appropriate postprocessing procedure, e.g., by filtering. However, within the context we use, i.e., direct minimization of functionals, we can perhaps more easily remove the oscillations by choosing a stronger functional; this approach is examined in Section 4.

4. COMPUTATIONAL EXPERIMENTS FOR MINIMUM $H^1(\Sigma)$ -NORM AND REGULARIZED MINIMUM $L^2(\Sigma)$ -NORM BOUNDARY CONTROLS

Exact minimum $L^2(\Sigma)$ -norm controls seemingly contain jump discontinuities; approximations contain oscillations and are thus not implementable in practice. We can remove both the jump discontinuities and the oscillations by requiring the solution to minimize a stronger functional.

In Sections 2 and 3, we focused on finding solutions of the controllability problem (1)–(3) that have minimum $L^2(\Sigma)$ -norm, i.e., that minimize the functional $\mathcal{J}_0(\cdot)$ of (8). We now focus on solutions that minimize the functional $\mathcal{J}_\epsilon(\cdot)$ of (10). Thus, instead of minimizing the discrete functional (29) which is an approximation to $\mathcal{J}_0(\Sigma)$, we now minimize a discrete approximation to the functional $\mathcal{J}_\epsilon(\Sigma)$; in this way we obtain an approximate solution of the controllability problem (12) and, by setting $\epsilon = 1$, of the controllability problem (11).

For the square domain $\Omega = (0, 1) \times (0, 1)$ and the time interval $(0, T)$, we have

$$\begin{aligned}
 \mathcal{J}_\epsilon(u) &\equiv \|u\|_{L^2(\Sigma)}^2 + \epsilon \|u\|_{H^1(\Sigma)}^2 \\
 &= \int_{\Sigma} u^2 d\Gamma dt + \epsilon \int_{\Sigma_1 \cap \Sigma_2} [u_t^2 + u_y^2] d\Gamma dt + \epsilon \int_{\Sigma_3 \cap \Sigma_4} [u_t^2 + u_x^2] d\Gamma dt \\
 &= \mathcal{J}_0(u) + \epsilon \sum_{k=0}^{K-1} \sum_{i=0}^{J-1} \int_{t_k}^{t_{k+1}} \int_{x_i}^{x_{i+1}} [u_t^2(z, 0, t) + u_t^2(z, 1, t) \\
 &\quad + (u_x^2(z, 0, t) + u_x^2(z, 1, t))] dz dt \\
 &\quad + \epsilon \sum_{k=0}^{K-1} \sum_{j=0}^{J-1} \int_{t_k}^{t_{k+1}} \int_{y_j}^{y_{j+1}} [u_t^2(0, z, t) + u_t^2(1, z, t) + u_y^2(0, z, t) + u_y^2(1, z, t)] dz dt.
 \end{aligned} \tag{43}$$

The corresponding discrete functional $\mathcal{J}_{\epsilon,h}(\cdot)$ is obtained using the trapezoidal rule for integration and second-order accurate difference quotient approximation for the derivatives, i.e., using $\mathcal{J}_{0,h}(\cdot)$

as given in (29) and

$$\begin{aligned} \int_{t_k}^{t_{k+1}} \int_{x_i}^{x_{i+1}} u_t^2(x, y_j, t) dz dt &\approx \frac{h}{2\delta} ((u_{i,j,k+1} - u_{i,j,k})^2 + (u_{i+1,j,k+1} - u_{i+1,j,k})^2), \\ \int_{t_k}^{t_{k+1}} \int_{x_i}^{x_{i+1}} u_x^2(x, y_j, t) dz dt &\approx \frac{\delta}{2h} ((u_{i+1,j,k} - u_{i,j,k})^2 + (u_{i+1,j,k+1} - u_{i,j,k+1})^2), \\ \int_{t_k}^{t_{k+1}} \int_{y_j}^{y_{j+1}} u_t^2(x_i, y, t) dz dt &\approx \frac{h}{2\delta} ((u_{i,j,k+1} - u_{i,j,k})^2 + (u_{i,j+1,k+1} - u_{i,j+1,k})^2), \\ \int_{t_k}^{t_{k+1}} \int_{y_j}^{y_{j+1}} u_y^2(x_i, y, t) dz dt &\approx \frac{\delta}{2h} ((u_{i,j+1,k} - u_{i,j,k})^2 + (u_{i,j+1,k+1} - u_{i,j,k+1})^2). \end{aligned}$$

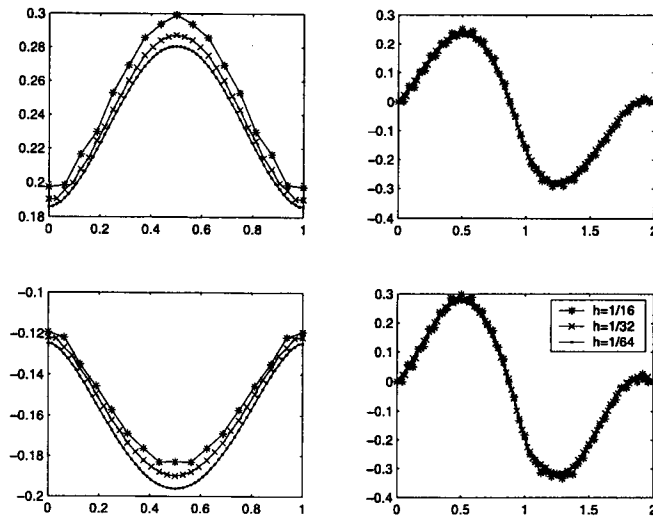


Figure 13. The approximation g^h that has minimum $H^1(\Sigma)$ -norm for the generic example with initial data (42)-I evaluated at $(0, y, T/4)$ (left-top); $(0, y, T/2)$ (left-bottom); $(1/4, 0, t)$ (right-top); and $(1/2, 0, t)$ (right-bottom).

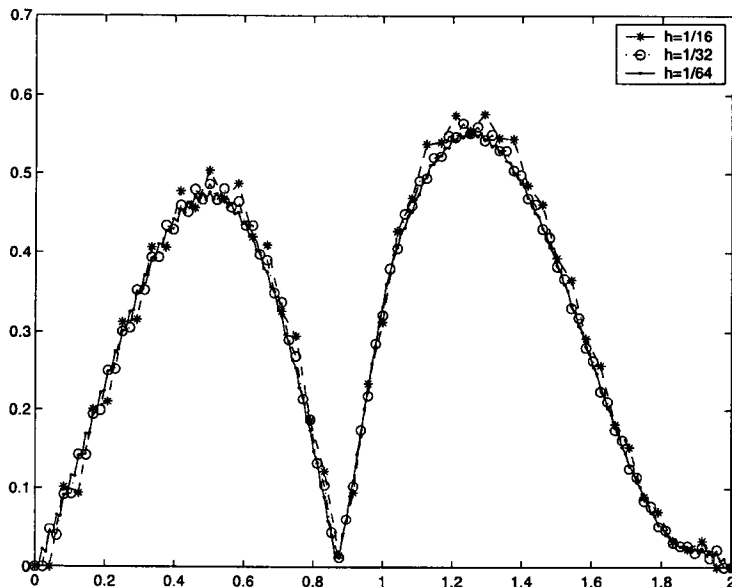


Figure 14. The norm $\|g^h(t)\|_{L_2(\Gamma)}$ of the approximate control that has minimum $H^1(\Sigma)$ -norm for the generic example with initial data (42)-I for $h = 1/16, 1/32$, and $1/64$.

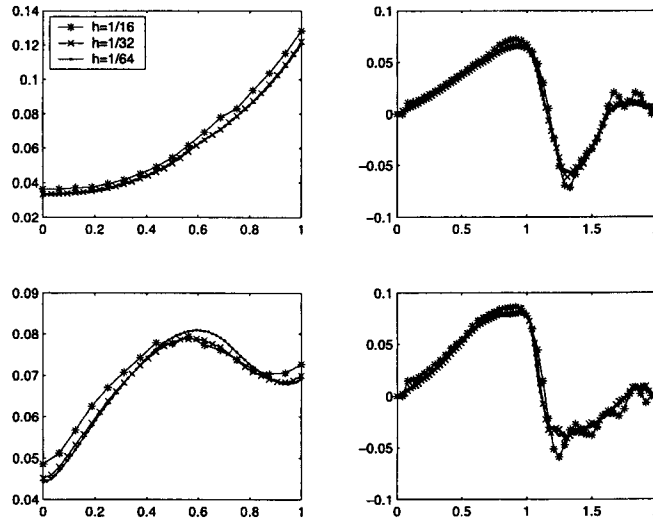


Figure 15. The approximation g^h that has minimum $H^1(\Sigma)$ -norm for the generic example with initial data (42)-II evaluated at $(0, y, T/4)$ (left-top); $(0, y, T/2)$ (left-bottom); $(1/4, 0, t)$ (right-top); and $(1/2, 0, t)$ (right-bottom).

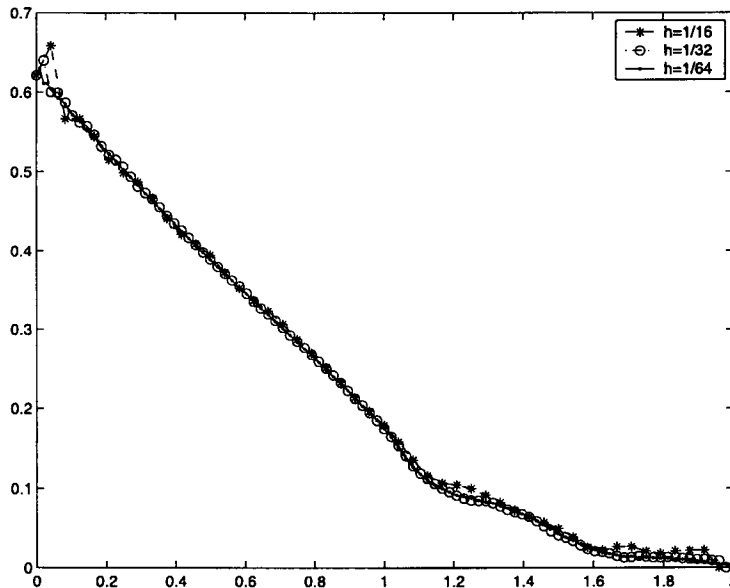


Figure 16. The norm $\|g^h(t)\|_{L_2(\Gamma)}$ of the approximate control that has minimum $H^1(\Sigma)$ -norm for the generic example with initial data (42)-II for $h = 1/16$, $1/32$, and $1/64$.

4.1. Approximations of Minimum $H^1(\Sigma)$ -Norm Controls

We set $\epsilon = 1$ and consider problem (11) so that we seek an approximation to minimum $H^1(\Sigma)$ -norm solutions of the controllability problem (1)–(3). Then, analogously to problem (30) for the minimum $L^2(\Sigma)$ -norm case, we are led to the following quadratic optimization problem:

$$\min_{\tilde{U} \in \mathbb{R}^{\tilde{q}}} \mathcal{J}_{1,h}(\tilde{U}), \quad \text{subject to } \tilde{A}_1 \tilde{U}_1 + \tilde{A}_2 \tilde{U}_2 = \tilde{F}, \quad \text{where } \mathcal{J}_{1,h}(\tilde{U}) = \tilde{U}_1 Q_1 \tilde{U}_1 + b,$$

with $Q_1 \in \mathbb{R}^{\tilde{q}_1 \times \tilde{q}_1}$ being a banded, positive definite matrix. The solution process is similar to what we have analyzed for discrete controls with minimum discrete $L^2(\Sigma)$ -norm in Section 2.

The results of computational experiments for generic examples with initial data (42) are presented in Table 3 and Figures 13–16. We compare Figures 13–16 with the previous figures for

Table 3. Results of the computational experiments for the minimum $H^1(\Sigma)$ -norm case for the generic examples with initial data (42).

	h	$\frac{1}{16}$	$\frac{1}{32}$	$\frac{1}{64}$
Case I	$\ g^h\ _{L^2(\Sigma)}$	0.5016	0.4875	0.4835
	$\ u^h\ _{L^2(Q)}$	0.5456	0.5308	0.5253
Case II	$\ g^h\ _{L^2(\Sigma)}$	0.4242	0.4215	0.4200
	$\ u^h\ _{L^2(Q)}$	0.2214	0.2185	0.2173

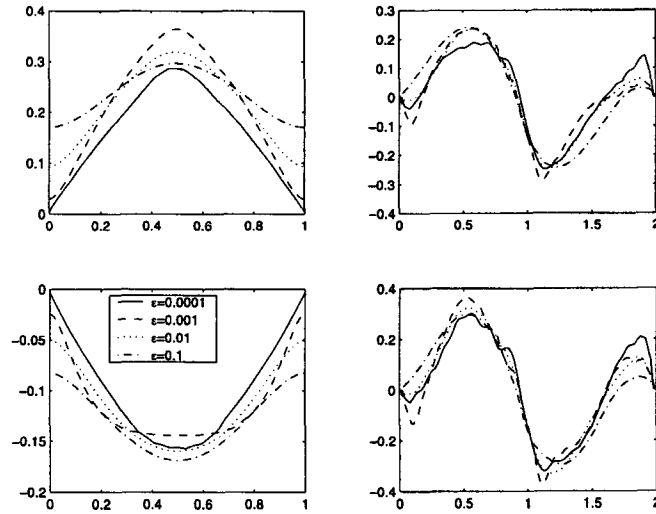


Figure 17. The approximation g^h that minimizes functional (44) for the generic example with initial data (42)-I for $h = 1/64$ and $\delta = 2h/3$ evaluated at $(0, y, T/4)$ (left-top); $(0, y, T/2)$ (left-bottom); $(1/4, 0, t)$ (right-top); and $(1/2, 0, t)$ (right-bottom).

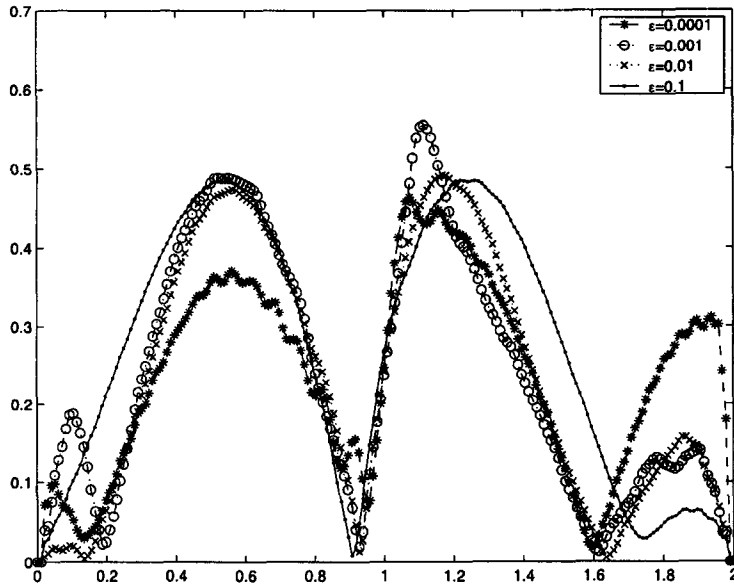


Figure 18. The norm $\|g^h(t)\|_{L_2(\Gamma)}$ of the approximate control that minimizes the functional (44) for the generic example with initial data (42)-I for $h = 1/64$ and $\delta = 2h/3$.

the minimum $L^2(\Sigma)$ -norm control case for which the same quantities are plotted for the same grid and initial data so that they differ only with regard to which discretized functional is being minimized. Clearly, we see that the solution obtained by minimizing the $H^1(\Sigma)$ -norm is much smoother than that obtained by minimizing the $L^2(\Sigma)$ -norm. Moreover, the pointwise convergence of the approximations to solutions of the minimum $H^1(\Sigma)$ -norm problem is evident, with an almost complete absence of oscillations.

4.2. Approximations of Regularized Minimum $L^2(\Sigma)$ -Norm Controls

Although they may be easier to obtain and apply and are more regular, minimum $H^1(\Sigma)$ -norm controls bear little resemblance to minimum $L^2(\Sigma)$ -norm controls. It may be the case, despite

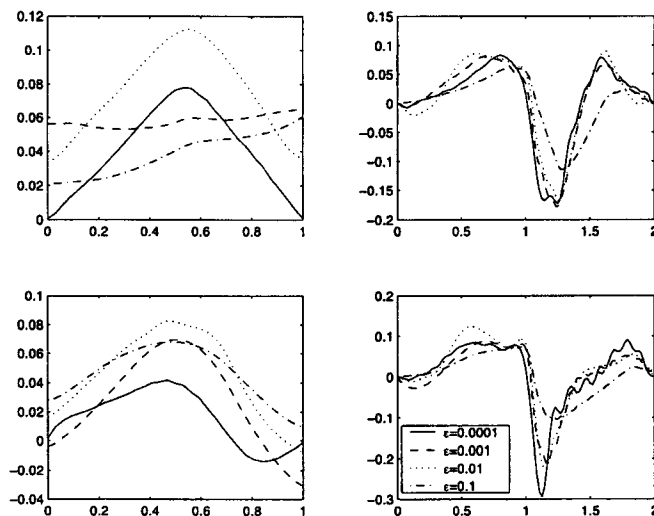


Figure 19. The approximation g^h that minimizes the functional (44) for the generic example with initial data (42)-II for $h = 1/64$ and $\delta = 2h/3$ evaluated at $(0, y, T/4)$ (left-top); $(0, y, T/2)$ (left-bottom); $(1/4, 0, t)$ (right-top); and $(1/2, 0, t)$ (right-bottom).

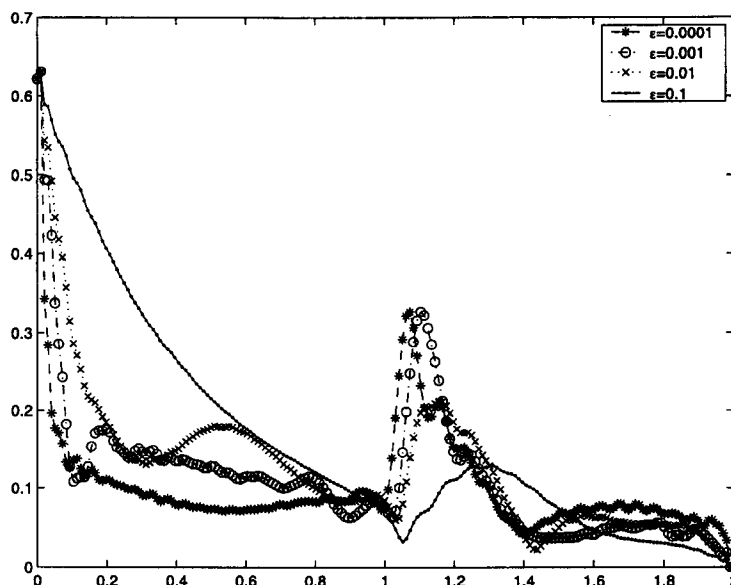


Figure 20. The norm $\|g^h(t)\|_{L_2(\Gamma)}$ of the approximate control that minimizes the functional (44) for the generic example with initial data (42)-II for $h = 1/64$ and $\delta = 2h/3$.

the fact that generically they are only piecewise smooth, that one really does wish to determine good approximations to the latter. To this end, we consider regularized minimum $L^2(\Sigma)$ -norm controls determined from a discretization of problem (12) of the form

$$\min_{\tilde{U} \in \mathbb{R}^{\tilde{q}}} \mathcal{J}_{\epsilon, h}(\tilde{U}), \quad \text{subject to } \tilde{A}_1 \tilde{U}_1 + \tilde{A}_2 \tilde{U}_2 = \tilde{F}, \quad \text{where } \mathcal{J}_{\epsilon, h}(\tilde{U}) = \tilde{U}_1 Q_\epsilon \tilde{U}_1 + b, \quad (44)$$

with $Q_\epsilon \in \mathbb{R}^{\tilde{q}_1 \times \tilde{q}_1}$ being a banded positive definite matrix. The solution process is similar to what we have analyzed for discrete controls with minimum discrete $L^2(\Sigma)$ or $H^1(\Sigma)$ -norms.

Computational results are given in Table 4 and in Figures 17–20, using the same computational parameters as were used previously for the generic examples with initial data (42). In Figures 21 and 22, we compare the approximate solutions obtained with $\epsilon = 0.0001$ with those obtained for $\epsilon = 0$.

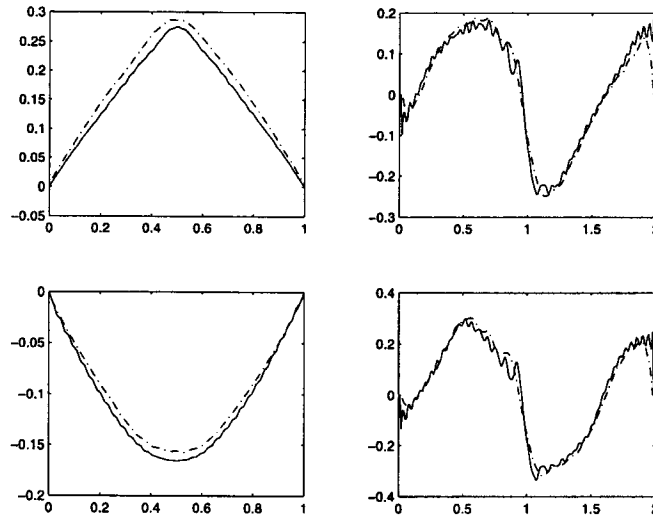


Figure 21. The approximate control g^h (—) that has minimum $L^2(\Sigma)$ -norm and the one $(\cdot - \cdot)$ that minimizes functional (44) with $\epsilon = 0.0001$ for the generic example with initial data (42)-I for $h = 1/64$ and $\delta = 2h/3$ evaluated at $(0, y, T/4)$ (left-top); $(0, y, T/2)$ (left-bottom); $(1/4, 0, t)$ (right-top); and $(1/2, 0, t)$ (right-bottom).

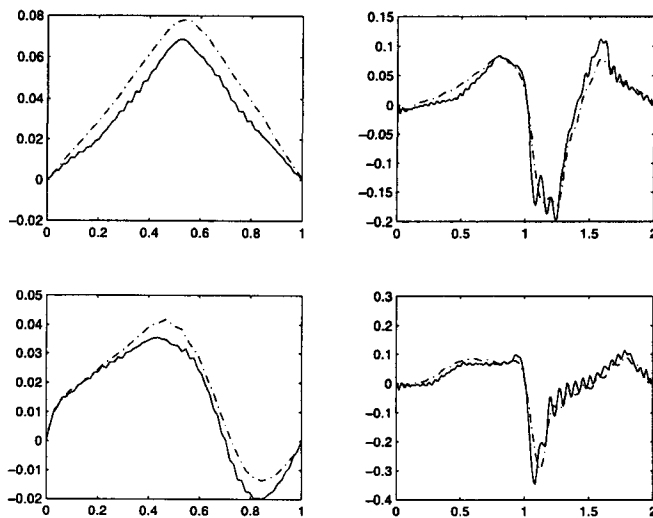


Figure 22. The approximate control g^h (—) that has minimum $L^2(\Sigma)$ -norm and the one $(\cdot - \cdot)$ that minimizes functional (44) with $\epsilon = 0.0001$ for the generic example with initial data (42)-II for $h = 1/64$ and $\delta = 2h/3$ evaluated at $(0, y, T/4)$ (left-top); $(0, y, T/2)$ (left-bottom); $(1/4, 0, t)$ (right-top); and $(1/2, 0, t)$ (right-bottom).

Table 4. Norms of the approximate controls and states that minimize functional (44) for the generic examples with initial data (42) for $h = 1/64$ and $\delta = 2h/3$.

		$\epsilon = 0$	$\epsilon = 0.0001$	$\epsilon = 0.001$	$\epsilon = 0.01$	$\epsilon = 0.1$	$\epsilon = 1$
Case I	$\ g^h\ _{L^2(\Sigma)}$	0.3850	0.3946	0.4018	0.4154	0.4461	0.4835
	$\ u^h\ _{L^2(Q)}$	0.5015	0.5051	0.4745	0.4887	0.5064	0.5253
Case II	$\ g^h\ _{L^2(\Sigma)}$	0.1753	0.1898	0.1971	0.2316	0.3096	0.4200
	$\ u^h\ _{L^2(Q)}$	0.1811	0.1803	0.1778	0.1791	0.1889	0.2173

From Figures 17–20, we see that, except for the smallest value of ϵ used ($\epsilon = 0.0001$), the approximate solutions obtained with $\epsilon > 0$ do not contain oscillations. From [15], we note that the oscillations for $\epsilon = 0.0001$ can be removed by reducing the grid sizes in time and space. Of course, regularized minimum $L^2(\Sigma)$ -norm controls become smoother and smoother as ϵ increases and as $\epsilon \rightarrow 1$, we recover the minimum $H^1(\Sigma)$ -norm case. From Table 4 and Figures 21 and 22, we see that the approximate controls converge, as $\epsilon \rightarrow 0$, to those obtained for $\epsilon = 0$. To summarize, the computational results indicate that by using a sufficiently small value of the penalty parameter and sufficiently small values of the spatial and temporal grid sizes h and δ , respectively, our methods produce pointwise convergent approximations to the exact controls, even when the latter are only piecewise smooth.

5. RELATIONSHIP OF NEW APPROACH AND HUM-BASED APPROACHES

For the minimum $L^2(\Sigma)$ -norm control setting, both our approach and HUM-based computational approaches [1,5,6,9–13] attempt to determine approximations that themselves minimize a discrete $L^2(\Sigma)$ -norm. Then, it is not surprising that the two methods are closely related. However, the computational results using the two approaches are very different, with convergent approximations being obtained with our methods whereas HUM-based approaches require regularization in order to produce stable and convergent approximations. Here, we explain the differences in the two approaches which apparently lead to the differences in performance.

The discrete equations of our approach and of HUM-based approaches for the minimum $L^2(\Sigma)$ -norm control setting are both subsets of a superset of equations which we write symbolically as

$$\begin{pmatrix} B_{\text{ini}}^u \\ B_{\text{ter}}^u \\ B_{\text{bdr}}^u \\ B_{\text{int}}^u \\ B_{\text{ini}}^s \\ B_{\text{ter}}^s \\ B_{\text{bdr}}^s \\ B_{\text{int}}^s \end{pmatrix} \begin{pmatrix} u_{\text{ini}} \\ u_{\text{ter}} \\ u_{\text{bdr}} \\ u_{\text{int}} \\ s_{\text{ini}} \\ s_{\text{ter}} \\ s_{\text{bdr}} \\ s_{\text{int}} \end{pmatrix} = \begin{pmatrix} U_{\text{ini}} \\ 0 \\ 0 \\ 0 \\ S_{\text{ini}} \\ S_{\text{ter}} \\ 0 \\ 0 \end{pmatrix}, \quad (45)$$

where the unknowns $u = \{u_{\text{ini}}, u_{\text{ter}}, u_{\text{bdr}}, u_{\text{int}}\}$ are nodal values of the state variable along the first two time levels, the last two time levels, the space-time boundary Σ , and the remaining time levels in the interior of Q , respectively, and likewise for the adjoint variables $s = \{s_{\text{ini}}, s_{\text{ter}}, s_{\text{bdr}}, s_{\text{int}}\}$. We also have the discrete operators:

- B_{ini}^u a discretization of the initial conditions for u that only involves us at first two time levels,
- B_{ter}^u a discretization of the terminal conditions for u that only involves us at last two time levels,
- B_{bdr}^u a discretization of the of the boundary relation $u = -\frac{\partial p}{\partial n}$ that relates boundary us to the ss ,

B_{int}^u the discrete wave equation operator applied to u ,

B_{ini}^s a discretization of the initial conditions for s that only involves ss at first two time levels,

B_{ter}^s a discretization of the terminal conditions for u that only involves ss at last two time levels,

B_{bdr}^s a discretization of the of the boundary relation $s = 0$ that only involves ss at the boundary,

B_{int}^s the discrete wave equation operator applied to s .

We have that U_{ini} is determined from the initial data of the problem, but S_{ini} and S_{ter} are not known. For example, we have $U_{\text{ini}} = U_0$ or $U_{\text{ini}} = U_0 + \Delta t U_1$, where U_0 and U_1 are the nodal values of the initial data for the state and for the derivative of the state. Part of $B_{\text{ini}}^u u = U_{\text{ini}}$ corresponds to the first time level in which case it simply states that $u_0 = U_0$. Part of $B_{\text{ini}}^u u = U_{\text{ini}}$ corresponds to the second time level; then it states that $u_1 = U_0 + \Delta t U_1$. Note that both U_0 and U_1 are known from the data of the problem. Similar relations hold for S_{ini} and S_{ter} , e.g., $B_{\text{ter}}^s s = S_{\text{ter}}$ is either $s_N = S_N$ or $s_{N-1} = S_N - \Delta t S_{N-1}$, where S_N and S_{N-1} are the nodal values of s and s_t at $t = T$. These values are not part of the specification of the problem and thus are unknown.

An examination of (45) reveals that it has the form

$$\begin{pmatrix} B_{\text{ini},1}^u & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & B_{\text{ter},2}^u & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & B_{\text{bdr},3}^u & 0 & B_{\text{bdr},5}^u & B_{\text{bdr},6}^u & B_{\text{bdr},7}^u & B_{\text{bdr},8}^u \\ B_{\text{int},1}^u & B_{\text{int},2}^u & B_{\text{int},3}^u & B_{\text{int},4}^u & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & B_{\text{ini},5}^s & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & B_{\text{ter},6}^s & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & B_{\text{bdr},7}^s & 0 \\ 0 & 0 & 0 & 0 & B_{\text{int},5}^s & B_{\text{int},6}^s & B_{\text{int},7}^s & B_{\text{int},8}^s \end{pmatrix} \begin{pmatrix} u_{\text{ini}} \\ u_{\text{ter}} \\ u_{\text{bdr}} \\ u_{\text{int}} \\ s_{\text{ini}} \\ s_{\text{ter}} \\ s_{\text{bdr}} \\ s_{\text{int}} \end{pmatrix} = \begin{pmatrix} U_{\text{ini}} \\ 0 \\ 0 \\ 0 \\ S_{\text{ini}} \\ S_{\text{ter}} \\ 0 \\ 0 \end{pmatrix}. \quad (46)$$

The subset of discrete equations for the HUM-based approach

In the HUM-based approach, the first and sixth equations from (46) are eliminated, resulting in (after a reordering of equations and unknowns)

$$\begin{pmatrix} B_{\text{ini},5}^s & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & B_{\text{bdr},7}^s & 0 & 0 & 0 & 0 & 0 & 0 \\ B_{\text{int},5}^s & B_{\text{int},7}^s & B_{\text{int},8}^s & B_{\text{int},6}^s & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & B_{\text{ter},2}^u & 0 & 0 & 0 \\ B_{\text{bdr},5}^u & B_{\text{bdr},7}^u & B_{\text{bdr},8}^u & B_{\text{bdr},6}^u & 0 & B_{\text{bdr},3}^u & 0 & 0 \\ 0 & 0 & 0 & 0 & B_{\text{int},2}^u & B_{\text{int},3}^u & B_{\text{int},4}^u & B_{\text{int},1}^u \end{pmatrix} \begin{pmatrix} s_{\text{ini}} \\ s_{\text{bdr}} \\ s_{\text{int}} \\ s_{\text{ter}} \\ u_{\text{ter}} \\ u_{\text{bdr}} \\ u_{\text{int}} \\ u_{\text{ini}} \end{pmatrix} = \begin{pmatrix} S_{\text{ini}} \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

or

$$\begin{pmatrix} B_{\text{ini},5}^s & 0 & 0 & 0 \\ 0 & B_{\text{bdr},7}^s & 0 & 0 \\ B_{\text{int},5}^s & B_{\text{int},7}^s & B_{\text{int},8}^s & B_{\text{int},6}^s \end{pmatrix} \begin{pmatrix} s_{\text{ini}} \\ s_{\text{bdr}} \\ s_{\text{int}} \\ s_{\text{ter}} \end{pmatrix} = \begin{pmatrix} S_{\text{ini}} \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad (47)$$

and

$$\begin{pmatrix} B_{\text{ter},2}^u & 0 & 0 & 0 \\ 0 & B_{\text{bdr},3}^u & 0 & 0 \\ B_{\text{int},2}^u & B_{\text{int},3}^u & B_{\text{int},4}^u & B_{\text{int},1}^u \end{pmatrix} \begin{pmatrix} u_{\text{ter}} \\ u_{\text{bdr}} \\ u_{\text{int}} \\ u_{\text{ini}} \end{pmatrix} = - \begin{pmatrix} 0 \\ B_{\text{bdr},5}^u s_{\text{ini}} + B_{\text{bdr},6}^u s_{\text{ter}} + B_{\text{bdr},7}^u s_{\text{bdr}} + B_{\text{bdr},8}^u s_{\text{int}} \\ 0 \end{pmatrix}. \quad (48)$$

System (47) corresponds to a discretization of (13) with S_{ini} in the former corresponding to e_0 and e_1 of the latter. System (48) corresponds to a discretization of (14). Given S_{ini} , (47) is uniquely solvable (by forward time marching) for $(s_{\text{ini}}, s_{\text{ter}}, s_{\text{bdr}}, s_{\text{int}})$. Once having determined $(s_{\text{ini}}, s_{\text{ter}}, s_{\text{bdr}}, s_{\text{int}})$, equation (48) is uniquely solvable (by backward time marching) for $(u_{\text{ini}}, u_{\text{ter}}, u_{\text{bdr}}, u_{\text{int}})$. Note that the given initial data U_{ini} does not appear in these equations and that the initial data S_{ini} is not known. Thus, these equations are solved by an iterative process: make a guess $S_{\text{ini}}^{(0)}$; then, for $k = 0, 1, 2, \dots$, first solve

$$\begin{pmatrix} B_{\text{ini},5}^s & 0 & 0 & 0 \\ 0 & B_{\text{bdr},7}^s & 0 & 0 \\ B_{\text{int},5}^s & B_{\text{int},7}^s & B_{\text{int},8}^s & B_{\text{int},6}^s \end{pmatrix} \begin{pmatrix} s_{\text{ini}}^{(k)} \\ s_{\text{bdr}}^{(k)} \\ s_{\text{int}}^{(k)} \\ s_{\text{ter}}^{(k)} \end{pmatrix} = \begin{pmatrix} S_{\text{ini}}^{(k)} \\ 0 \\ 0 \end{pmatrix}$$

for $(s_{\text{ini}}^{(k)}, s_{\text{ter}}^{(k)}, s_{\text{bdr}}^{(k)}, s_{\text{int}}^{(k)})$; then, solve

$$\begin{pmatrix} B_{\text{ter},2}^u & 0 & 0 & 0 \\ 0 & B_{\text{bdr},3}^u & 0 & 0 \\ B_{\text{int},2}^u & B_{\text{int},3}^u & B_{\text{int},4}^u & B_{\text{int},1}^u \end{pmatrix} \begin{pmatrix} u_{\text{ter}}^{(k)} \\ u_{\text{bdr}}^{(k)} \\ u_{\text{int}}^{(k)} \\ u_{\text{ini}}^{(k)} \end{pmatrix} = - \begin{pmatrix} 0 \\ B_{\text{bdr},5}^u s_{\text{ini}}^{(k)} + B_{\text{bdr},6}^u s_{\text{ter}}^{(k)} + B_{\text{bdr},7}^u s_{\text{bdr}}^{(k)} + B_{\text{bdr},8}^u s_{\text{int}}^{(k)} \\ 0 \end{pmatrix},$$

for $(u_{\text{ini}}^{(k)}, u_{\text{ter}}^{(k)}, u_{\text{bdr}}^{(k)}, u_{\text{int}}^{(k)})$. (Note that, in general, we have that the initial conditions for u are not satisfied, i.e., $B_{\text{int}}^u u^{(k)} \neq U_{\text{ini}}$.) Then, one determines the update

$$S_{\text{ini}}^{(k+1)} = S_{\text{ini}}^{(k)} + \delta S_{\text{ini}}^{(k)},$$

where $\delta S_{\text{ini}}^{(k)}$ is determined through the HUM process, i.e., viewing (47) and (48) as implicitly defining a mapping (a discretization of (15)) between S_{ini} and U_{ini} and then using that mapping to define the update δS_{ini} .

The subset of discrete equations for the new approach

Our approach corresponds to *eliminating the equations in (46) that involve unknown data*, i.e., those involving S_{ini} and S_{ter} , resulting in

$$\begin{pmatrix} B_{\text{ini},1}^u & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & B_{\text{ter},2}^u & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & B_{\text{bdr},3}^u & 0 & B_{\text{bdr},5}^u & B_{\text{bdr},6}^u & B_{\text{bdr},7}^u & B_{\text{bdr},8}^u \\ B_{\text{int},1}^u & B_{\text{int},2}^u & B_{\text{int},3}^u & B_{\text{int},4}^u & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & B_{\text{bdr},7}^s & 0 \\ 0 & 0 & 0 & 0 & B_{\text{int},5}^s & B_{\text{int},6}^s & B_{\text{int},7}^s & B_{\text{int},8}^s \end{pmatrix} \begin{pmatrix} u_{\text{ini}} \\ u_{\text{ter}} \\ u_{\text{bdr}} \\ u_{\text{int}} \\ s_{\text{ini}} \\ s_{\text{ter}} \\ s_{\text{bdr}} \\ s_{\text{int}} \end{pmatrix} = \begin{pmatrix} U_{\text{ini}} \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

Given U_{ini} , this system is uniquely solvable. One can, for instance, solve this system by a direct method to obtain an approximate solution of the minimum $L^2(\Sigma)$ -norm exact controllability problem.

In summary, we see that both the HUM-based and our approaches deal with the same superset of discrete equations (46), but that this set of equations is solved by different means. Thus, it is apparent that the way one solves (46) accounts for the differences in the performance of the two types of approaches.

6. CONCLUDING REMARKS

We highlight again some of the main contributions of this paper in numerical solutions of controllability problems for the wave equation.

- For minimum L^2 -norm control, our method gives pointwise convergent solutions when the exact minimum L^2 -norm control is smooth; it gives L^2 convergent solutions when the exact solution is nonsmooth (e.g., the exact minimum L^2 norm control on the boundary is piecewise smooth with jump discontinuities). Furthermore, convergence is achieved for arbitrary time/space mesh size ratios satisfying the CFL condition.
- For minimum H^1 -norm control, our method gives pointwise convergent solutions. Also, jump discontinuities are ruled out by the H^1 norm.
- For minimum regularized L^2 -norm control, our method gives pointwise convergent solutions with mesh refinement. As the penalty constant tends to 0, the solution of the penalized problem converges in L^2 to the minimum L^2 -norm solution and becomes more and more oscillatory near jump discontinuities in the generic case.

In our approach, the boundary condition $u|_{\Sigma} = g$ is not applied; one merely allows for the discrete approximations to u along Σ to be included in the list of unknowns. Once we have obtained an approximation u^h to u by solving a discretized optimization problem such as (30), we can use that approximation to define any type of control we wish. If we set $g^h = u^h|_{\Sigma}$, we, of course, define an approximate Dirichlet control. We could just as well set $g^h = D_h u^h$, where D_h is a finite-difference quotient operator approximating the normal derivative along Γ ; in this case we define an approximate Neumann control along Σ . We may, in fact, define any type of linear or nonlinear control in this manner. However, the particular solution we use may not be minimizing the control function. For example, if we solve the optimization problem (30) and then set $g^h = D_h u^h$ we are tacitly applying Neumann control but we are minimizing an approximation to the $L^2(\Sigma)$ -norm of u (and not of $g = (\partial_n u)|_{\Sigma}$).

We note that the computational examples presented here and in [15] do not treat the case of problems with initial data for the solution function having jump discontinuities, i.e., either we have treated smooth initial data, or, as in Section 3.1, problems for which the initial data for the time derivative of the solution has jump discontinuities. In cases for which the initial data for the solution function has jump discontinuities, our methods still yield different results from HUM-based methods. For the latter methods, one obtains unstable approximations and regularization of the discrete equations is necessary in order to obtain convergent approximations; see [13]. For our methods, we obtain stable but nonconvergent approximations; convergent approximations can be obtained either by regularizing the data or by a postprocessing averaging procedure. These types of regularizations are fundamentally different from those necessary for HUM-based approaches. Details will be given in [17].

In future work, we will explore in greater detail the use, effect, and practical consequences of different functionals in exact controllability problems for the wave equation. We will also study the approximate solution of such problems by our methods in the contexts of both finite difference and finite-element methods and we will develop efficient solution strategies for the discrete problems. We will also apply our methodology to other problems, most notably the equations of linear elasticity.

Finally, we note that, at this point, we do not have a rigorous convergence proof for our method; our conclusions on the seeming convergence properties of our method are based on computational experiments and comparisons with results obtained using other methods.

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